

EEE 402 (January 2026)
Artificial Intelligence and Machine Learning Laboratory

Final Project Report

Section: G2 Group: 13

Telling Lung Adenocarcinoma from Squamous Cell Carcinoma on CT Using Hospital-Robust Radiomics

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1 Abstract

Adenocarcinoma (ADC) and squamous cell carcinoma (SCC) are the two main types of non-small cell lung cancer, and they are treated differently. Today the type is confirmed by biopsy, which is invasive, while almost every patient already has a chest CT scan. In this project we built a fully automatic CT pipeline that predicts ADC versus SCC, and we designed its evaluation so that the result holds at hospitals the model has never seen.

We first showed that pooling CT data from several hospitals creates a shortcut: hospitals differ both in scanner style and in their ADC/SCC mix, so a predictor that only knows the hospital name reaches a pooled ROC-AUC of 0.71 without looking at any image. To remove this effect we used a site-balanced AUC, ComBat harmonisation fitted inside every training fold, and leave-one-hospital-out testing. The tumour is located by an expert box or TotalSegmentator and outlined by MedSAM2. Texture features are extracted from three peritumoral shells (0-4, 4-8 and 8-12 mm) and combined with tumour position, size, age and sex. Two feature blocks are each modelled by an ensemble of logistic regression, SVM, LightGBM and ridge regression, and the block scores are fused.

On 1041 patients from four public cohorts (697 ADC, 344 SCC) the model reached a site-balanced AUC of 0.722 (95% CI 0.687-0.762) in repeated nested cross-validation and 0.714 on average for an unseen hospital. In two locked, pre-registered external tests it scored 0.686 on Lung3 (79 patients) and 0.585 on TCGA (84 patients). Thin-slice CT gave the best external results, while 5 mm CT was the main failure mode. Larger deep models (attention-MIL, CNN features, CT foundation models and adversarial fine-tuning) did not beat the simpler harmonised radiomics model at this data size.

Keywords: lung cancer, histologic subtype, radiomics, ComBat harmonisation, multi-centre validation, site-balanced AUC, MedSAM2.

2 Introduction

Lung cancer is the leading cause of cancer death worldwide [1]. About 85% of cases are non-small cell lung cancer (NSCLC), and its two most common histologic types are adenocarcinoma (ADC) and squamous cell carcinoma (SCC) [2]. Knowing the type matters because treatment depends on it. Some chemotherapy drugs are used for one type and not the other, and most targeted therapies (for example for EGFR and ALK changes) are relevant mainly to ADC. At present the type is confirmed by a biopsy: a needle through the chest wall, a bronchoscope, or surgery. A biopsy is invasive, carries risks such as collapsed lung and bleeding, and sometimes the sample is too small to decide.

Almost every lung cancer patient has a chest CT scan before any treatment. On CT, ADC and SCC show different tendencies. ADC is more often peripheral and may have fuzzy or ground-glass edges, while SCC is more often central, solid and larger, and is more common in men and heavy smokers. These are tendencies and not rules, so the two types often look alike. *Radiomics* tries to capture such differences by computing many quantitative features (intensity, texture and shape) from the tumour region and feeding them to a machine-learning model [3, 4]. Our project question was therefore: **can a computer tell ADC from SCC using only the CT scan, and does the answer still hold at hospitals that were not used for training?**

The second half of this question became the centre of the project. Public CT collections come from different hospitals, and each hospital has its own scanners and its own mix of patients. In our data one hospital is 75% SCC while the other three are 73% to 81% ADC. A model trained on the pooled data can learn to recognise the hospital from scanner texture and then guess the majority class of that hospital. This is a well-known form of *shortcut learning* [5, 6]. It gives a good pooled score but fails at a new hospital. A large part of this work was spent on measuring this shortcut and on building an evaluation that it cannot fool.

Objectives. The objectives of the project were:

1. to build a fully automatic CT pipeline (tumour detection, outlining, feature extraction and classification) for ADC versus SCC;
2. to measure the hospital shortcut and use a metric and validation design that are not affected by it;
3. to reduce scanner differences between hospitals by harmonisation, fitted on training data only;
4. to compare simple handcrafted models with deep-learning alternatives under the same honest protocol;
5. to test the frozen model once on two external datasets that were locked until the end.

Why this is a Complex Engineering Problem. Table 1 maps the project to the attributes of a complex engineering problem. The main difficulty is that there is no obvious solution: a higher score on pooled data can be a worse model, so success itself had to be defined carefully.

Attribute	How it appears in this project
P1: Depth of knowledge	Needs CT physics (Hounsfield units, slice thickness, reconstruction kernels), image processing (resampling, segmentation), statistics (empirical Bayes, bootstrap) and machine learning.
P2: Conflicting requirements	Higher pooled accuracy conflicts with fairness across hospitals; removing scanner effects can also remove real disease signal; model capacity conflicts with small data.
P3: Depth of analysis	No closed-form answer. Many design choices (regions, features, harmonisation, models, fusion) had to be tested in nested cross-validation without leaking test information.
P4: Familiarity of issues	Hidden confounding between hospital and label is not visible in standard metrics and is rarely checked in the literature.
P5: Extent of applicable codes	Radiomics standards (IBSI), DICOM, data licences (CC BY, CC BY-NC) and good practice for medical AI evaluation (pre-registration, locked test sets).
P6: Stakeholders	Patients, radiologists, pathologists, oncologists and hospitals with different scanners.
P7: Interdependence	Mask quality affects features, features affect harmonisation, harmonisation affects model selection; a change in one stage changes all later stages.

Table 1: Mapping of the project to the attributes of a complex engineering problem

Possible alternative solutions. Several other routes could address the same clinical need. Table 2 compares them with the approach we chose.

Alternative	Advantage	Disadvantage
Biopsy only (current practice)	Gold standard; also gives tissue for gene tests	Invasive, risky, sometimes not diagnostic, slow
Visual reading by a radiologist	No new tools needed	Subjective; ADC and SCC often look alike
End-to-end deep CNN on CT	Learns features automatically	Needs thousands of patients; tends to learn scanner style (shown in Section 5.3)
Single-hospital radiomics	Simple, high internal scores	Rarely transfers to other hospitals
PET/CT based model	Uses metabolic information	PET is not available for every patient
Multi-hospital harmonised radiomics (chosen)	Works with routine CT, small models, honest multi-centre evaluation	Moderate accuracy; sensitive to thick CT slices

Table 2: Alternative solutions considered

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

The scope of the project was fixed after the progress presentation (Week 10). **In scope:**

- binary classification of ADC versus SCC from a pre-treatment chest CT scan and the patient's age and sex;
- public, de-identified CT collections from The Cancer Imaging Archive (TCIA) [7] and the NCI Imaging Data Commons (IDC) [8];
- fully automatic tumour localisation and outlining, so that the same method works at a new hospital;
- evaluation inside the development hospitals (cross-validation), on unseen development hospitals (leave-one-hospital-out) and on two locked external datasets.

Out of scope: other NSCLC types (large cell, not otherwise specified), PET images, gene mutation prediction, survival prediction, and any clinical deployment. The model is a research decision-support tool, not a replacement for biopsy.

3.1.2 Literature Review

Radiomics for histologic subtype. Zhu *et al.* [9] used five LASSO-selected radiomic features on 129 patients from one hospital and reported an AUC of 0.91, but without external testing. Yang *et al.* [10] studied 645 patients from three centres. When centres were mixed randomly the AUC was about 0.78, but when the model was trained on one centre and tested on another it fell to 0.54-0.64. Pasini *et al.* [11] used LUNG1 and NSCLC-Radiogenomics (two of our cohorts) and found that accuracy fell from 0.77 to 0.59 after ComBat harmonisation, which suggests that part of the pooled performance came from dataset differences. Song *et al.* [12] collected 868 patients from eight TCIA databases and reported AUCs of 0.82 internally and 0.82 and 0.80 on TCGA and Lung3, choosing the best of 130 radiomics models and using expert tumour masks.

Deep learning. Chaunzwa *et al.* [13] trained a CNN on 311 patients from one hospital (internal AUC 0.71) and obtained 0.60 on an external Lung3 subset. Chen *et al.* [14] used a multi-task CNN on 402 TCIA patients (internal 0.84, external 0.73). Foundation models pre-trained on large CT collections, such as FMCIB [15] and CT-FM [16], give general-purpose image features that can be reused without training a network from scratch.

Peritumoral and local information. Tang *et al.* [17] showed that features from the region around the tumour add information to those from inside it. Vuong *et al.* [18] used local radiomic activation maps to see where the signal lies inside and around the tumour. Weakly supervised attention-based multiple instance learning (MIL) [19, 20] learns from many local patches when only a patient-level label exists; this was the starting idea of our project.

Multi-centre effects. ComBat [21] was designed to remove batch effects in gene expression data and has been adapted to radiomics [22]. Shortcut learning of hospital identity was documented in chest radiographs [5] and in deep networks in general [6]. Radiomic feature definitions are standardised by the IBSI [23], but standard definitions do not remove scanner differences.

Gap. Most published ADC/SCC models report high scores from one hospital or from random splits of pooled data. Very few measure how much of the score is explained by the hospital itself, and very few test a model that was frozen before the external labels were opened. Our project addresses this gap.

3.1.3 Formulation of Problem

We have N patients from H hospitals. Patient i has a CT volume I_i , a hospital label $h_i \in \{1, \dots, H\}$ and a histology label $y_i \in \{0,1\}$, where $y_i = 1$ means ADC (the positive class) and $y_i = 0$ means SCC. We want a function f that maps a CT scan to a real-valued score $s_i = f(I_i)$, with $\hat{y}_i = \mathbb{1}[s_i > 0]$. Ranking quality is measured by the ROC-AUC, which is the probability that a random ADC patient receives a higher score than a random SCC patient:

$$\text{AUC} = \frac{1}{n_1 n_0} \sum_{i: y_i=1} \sum_{j: y_j=0} [\mathbb{1}(s_i > s_j) + \frac{1}{2} \mathbb{1}(s_i = s_j)] \quad (1)$$

where n_1 and n_0 are the numbers of ADC and SCC patients. The real goal is not the AUC on pooled data but the AUC at a hospital h^* that was not used for training:

$$f^* = \underset{f}{\operatorname{argmax}} \mathbb{E}_{h^* \notin \mathcal{H}_{\text{train}}} [\text{AUC}_{h^*}(f)] \quad (2)$$

From this goal we derived the following design requirements:

- **R1.** The metric used for model selection must give 0.5 to any predictor that only knows the hospital.
- **R2.** Every fitted step (imputation, harmonisation, scaling, feature selection, hyperparameters) must use training patients only.
- **R3.** Tumour masks must be produced by the same automatic method at every hospital, so that the mask source cannot become a new hospital fingerprint.
- **R4.** A new hospital must be processed without its labels.
- **R5.** The model must be small enough to train on a laptop GPU (4 GB) and to avoid overfitting about 1000 patients.
- **R6.** External tests must be pre-registered, frozen with a hash and scored once.

3.1.4 Analysis

Figure 1 shows the ADC/SCC mix of every cohort. LUNG1 is 25% ADC, whereas Radiogenomics, Lung-PET-CT-Dx and NLST are 79%, 81% and 73% ADC. Consider a predictor that never looks at the image and outputs only the ADC rate of the patient's hospital:

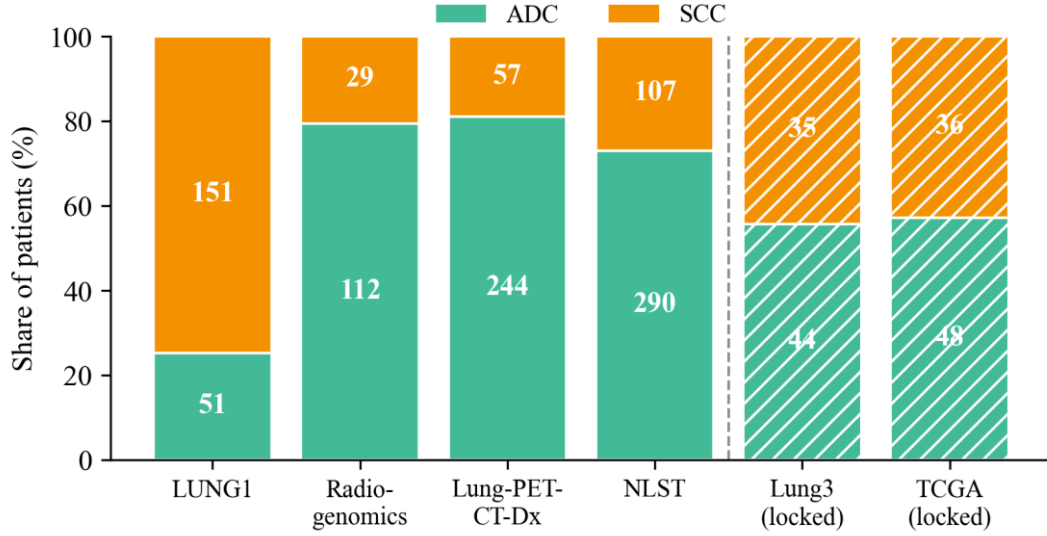


Figure 1: Class mix of the four development cohorts and the two locked external test sets. Numbers inside the bars are patient counts. LUNG1 is reversed compared with the other hospitals.

$$s_i = \pi_{h_i}, \pi_h = \frac{n_{h,1}}{n_{h,1} + n_{h,0}} \quad (3)$$

Every ADC patient from a high-ADC hospital is ranked above every SCC patient from LUNG1, so this predictor obtains a pooled AUC of **0.710** on our 1041 patients (Figure 2). A naive radiomics model scores 0.727, which is only slightly better than knowing the hospital. A pooled AUC around 0.7 therefore says very little about whether a model has learned anything about the tumour.

To satisfy requirement R1 we weight each patient by the inverse size of its (hospital, class) group:

$$w_i = \frac{1}{H \cdot n_{h_i, y_i}} \quad (4)$$

and define the **site-balanced AUC** as the weighted version of Eq. (1):

$$\text{AUC}_{\text{SB}} = \frac{\sum_{i: y_i=1} \sum_{j: y_j=0} w_i w_j [\mathbb{1}(s_i > s_j) + \frac{1}{2} \mathbb{1}(s_i = s_j)]}{\sum_{i: y_i=1} w_i \sum_{j: y_j=0} w_j} \quad (5)$$

After weighting, each of the $2H$ (hospital, class) groups has the same total weight $1/H$. For the hospital-only predictor of Eq. (3), pairs from the same hospital are ties and contribute $1/2$. For two different hospitals h and h' , the ordered pairs (ADC from h , SCC from h') and (ADC from h' , SCC from h) together contribute exactly 1, whichever of π_h and $\pi_{h'}$ is larger. Hence

$$\text{AUC}_{\text{SB}}(\pi) = \frac{1}{H^2} \left[\sum_{h=1}^H \frac{1}{2} + \sum_{h < h'} 1 \right] = \frac{1}{H^2} \left[\frac{H}{2} + \frac{H(H-1)}{2} \right] = \frac{1}{2} \quad (6)$$

so the shortcut is worth exactly nothing under this metric, as required. On real data the hospital-only predictor scores 0.497 and the naive radiomics model 0.642 (Figure 2). The same weights w_i are used as sample weights during training, so the classifiers are not rewarded for predicting the hospital either. The second part of the analysis concerns scanner style: a logistic regression trained to recognise the hospital from texture features reached an AUC of 0.81-0.91 (Section 5.3.3). This motivated harmonisation (Section 3.2.4).

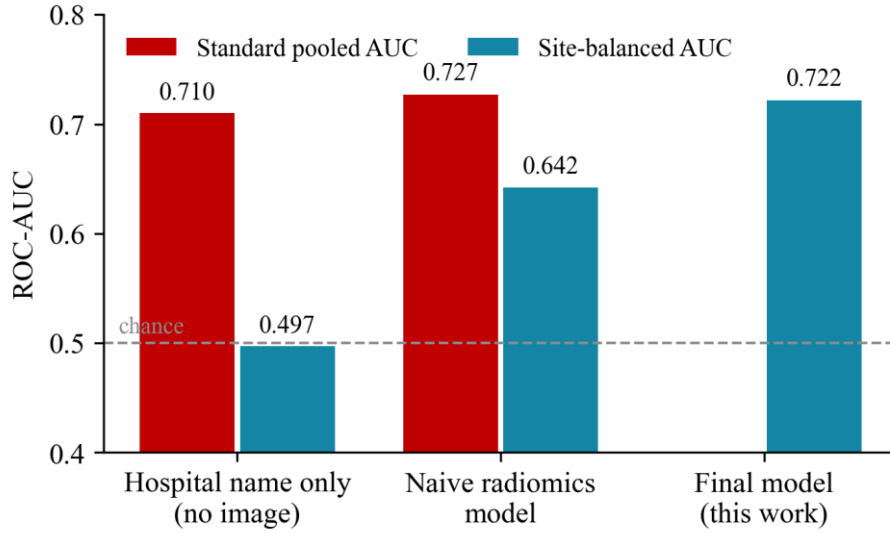


Figure 2: The hospital shortcut. A predictor that only knows the hospital reaches a pooled AUC of 0.710 but exactly chance under the site-balanced AUC. The final model reaches 0.722 under the fair metric.

3.2 Design Method (PO(a))

This section gives the mathematics behind every stage of the pipeline. The overall flow is shown later in Figure 4.

3.2.1 Image Preprocessing

Each DICOM series is stacked into one 3-D volume and converted to NIfTI. Scans have pixel sizes of 0.6-1 mm and slice gaps of 0.6-10 mm, so all volumes are resampled to an isotropic grid of $2 \times 2 \times 2$ mm. The CT is interpolated linearly: the value at a new voxel position x is a weighted average of the 8 surrounding original voxels,

$$I'(x) = \sum_{k=1}^8 w_k(x) I(x_k), \sum_{k=1}^8 w_k(x) = 1 \quad (7)$$

where the weights are products of the fractional distances along each axis. Masks are resampled by nearest neighbour so that they stay binary. Intensities are then clipped to the range from air to soft tissue, and grey levels are grouped into bins of width $\Delta = 20$ HU before texture is computed:

$$\tilde{I}(x) = \min(\max(I'(x), -1024), 200), b(x) = \lfloor \frac{\tilde{I}(x) + 1024}{\Delta} \rfloor + 1 \quad (8)$$

Clipping removes the effect of bone and metal, and binning makes texture less sensitive to small noise. 2 mm was chosen as a compromise: many scans have 3-5 mm slices, and a finer grid would only invent detail.

3.2.2 Tumour Segmentation and Peritumoral Shells

For development cohorts the tumour location is an expert box (LUNG1 and Radiogenomics: the bounding box of the expert outline; Lung-PET-CT-Dx and NLST: expert boxes). For external data the box comes from the TotalSegmentator lung-nodule model [24], taking the largest detected nodule. The box is given as a prompt to MedSAM2 [25], a medical version of the Segment Anything 2 model. MedSAM2 treats the axial slices as frames of a video: it outlines the tumour on the key slice and propagates the outline up and down. Segmentation quality against the expert outline B is measured by the Dice score,

$$\text{DSC}(A, B) = \frac{2|A \cap B|}{|A| + |B|} \quad (9)$$

which is 1 for identical masks and 0 for no overlap. From the tumour mask G we compute the Euclidean distance of every voxel to the tumour and build three shells around it, with radii $(r_0, r_1, r_2, r_3) = (0, 4, 8, 12)$ mm:

$$d(x) = \min_{g \in G} \|x - g\|_2, S_k = \{x: r_{k-1} < d(x) \leq r_k\} \quad (10)$$

The shells describe the tumour border and its surroundings, where ADC and SCC are expected to differ [17].

3.2.3 Radiomic and Position Features

Features are computed with PyRadiomics [4] on the original image, following IBSI definitions [23]. Each region gives 93 texture features from six families: first-order statistics, GLCM, GLRLM, GLSZM, GLDM and NGTDM. Three examples show the idea. If $p(i)$ is the fraction of voxels in bin i , $p(i, j)$ the normalised grey-level co-occurrence matrix and $P(i, j)$ the number of runs of grey level i with length j , then

$$H = - \sum_{i=1}^{N_g} p(i) \log_2(p(i) + \epsilon), \text{Contrast} = \sum_{i=1}^{N_g} \sum_{j=1}^{N_g} (i - j)^2 p(i, j) \quad (11)$$

$$\text{SRE} = \frac{\sum_i \sum_j P(i, j) / j^2}{\sum_i \sum_j P(i, j)} \quad (12)$$

Entropy H measures randomness of intensity, GLCM contrast measures local intensity jumps, and short run emphasis (SRE) is large when the texture is fine. The final model uses two feature blocks:

- **Shells block (279 features):** 93 texture features in each of the three shells.
- **Position + clinical block (13 features):** left or right lung; relative lateral, front-back and up-down position of the tumour centre; normalised distance from the midline; depth of the centre (in mm and relative); fraction of the tumour surface touching the lung boundary and the medial (mediastinal) part of it; fraction of the tumour outside the lung; log tumour volume; age; sex.

Other blocks were tested as alternatives: whole-tumour texture plus 14 shape features (GTV, 107 features), an inner 4 mm ring inside the tumour, FMCIB deep features [15] and CT-FM features [16].

3.2.4 Harmonisation with ComBat

ComBat [21] assumes that feature g of patient i in hospital h is shifted and scaled by the hospital:

$$x_{ihg} = \alpha_g + \sigma_g \gamma_{hg} + \sigma_g \sqrt{\delta_{hg}} \varepsilon_{ihg}, \varepsilon_{ihg} \sim \mathcal{N}(0, 1) \quad (13)$$

where α_g is the overall mean, σ_g the pooled standard deviation, γ_{hg} the hospital's location shift and δ_{hg} its scale factor. After standardising with the pooled estimates, the raw shift and scale of each hospital are

$$z_{ihg} = \frac{x_{ihg} - \hat{\alpha}_g}{\hat{\sigma}_g}, \hat{\gamma}_{hg} = \frac{1}{n_h} \sum_{i \in h} z_{ihg}, \hat{\delta}_{hg} = \frac{1}{n_h} \sum_{i \in h} (z_{ihg} - \hat{\gamma}_{hg})^2 \quad (14)$$

Small hospitals give noisy estimates, so ComBat uses empirical Bayes: across all features of hospital h it assumes $\gamma_{hg} \sim \mathcal{N}(\bar{\gamma}_h, \bar{\tau}_h^2)$ and $\delta_{hg} \sim \text{InvGamma}(\lambda_h, \theta_h)$, estimates these priors from the data, and iterates the posterior means

$$\gamma_{hg}^* = \frac{n_h \bar{\tau}_h^2 \gamma_{hg} + \delta_{hg}^* \bar{\gamma}_h}{n_h \bar{\tau}_h^2 + \delta_{hg}^*}, \delta_{hg}^* = \frac{\frac{1}{2} \sum_{i \in h} (z_{ihg} - \gamma_{hg}^*)^2 + \theta_h}{\frac{n_h}{2} + \lambda_h - 1} \quad (15)$$

until they converge. The harmonised feature is

$$x_{ihg}^* = \hat{\sigma}_g \frac{z_{ihg} - \gamma_{hg}^*}{\sqrt{\delta_{hg}^*}} + \hat{\alpha}_g \quad (16)$$

ComBat is fitted on the training patients of each fold only. A hospital that was not in the training data (an unseen or external hospital) gets its own $\hat{\gamma}$ and $\hat{\delta}$ from its *unlabelled* features with Eq. (14), so no test label is used. Figure 3 shows the effect on one real feature: before ComBat the LUNG1 curve is clearly shifted, after ComBat the four hospitals overlap.

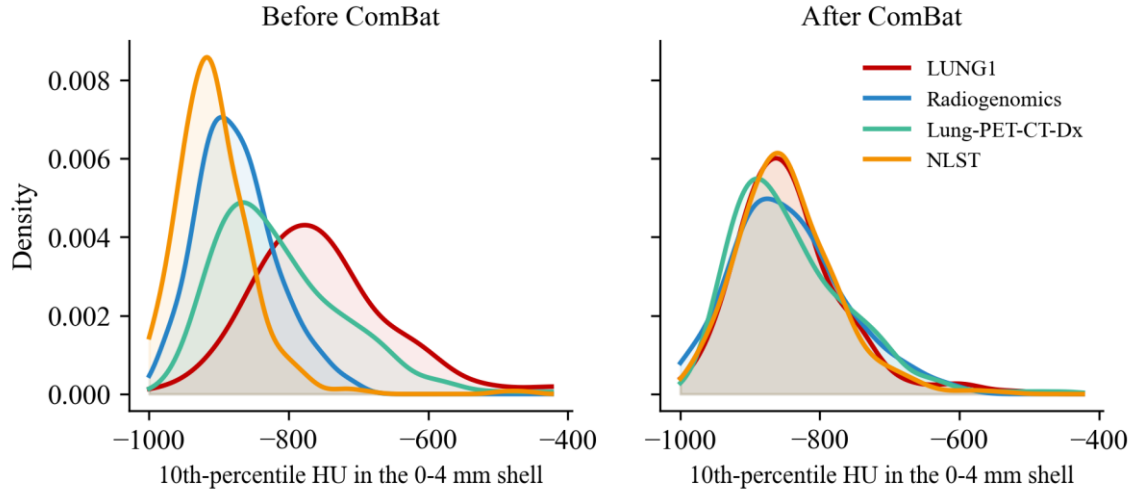


Figure 3: Effect of ComBat on one shell feature (10th-percentile HU in the 0-4 mm shell) for all 1041 patients. Left: each hospital has its own location and spread. Right: after ComBat the distributions are aligned. (Fitted on all patients for illustration only.)

3.2.5 Feature Scaling, Ranking and Selection

Inside each training fold, features with more than 20% missing values or zero variance are dropped and missing values are replaced by the training median. After ComBat every feature is z-scored with the training mean and standard deviation, $\tilde{x}_{ig} = (x_{ig}^* - \mu_g^{\text{tr}})/s_g^{\text{tr}}$. Features are then ranked by how well they separate ADC from SCC *within* each hospital:

$$\rho_g = \frac{1}{H} \sum_{h=1}^H |\text{AUC}_h(\tilde{x}_g) - 0.5| \quad (17)$$

where AUC_h is computed with Eq. (1) on hospital h only. Ranking within hospitals avoids choosing features that mainly separate hospitals. Going down the ranked list, a feature is skipped if its absolute correlation with an already kept feature is above 0.9, $|r_{gk}| = |\frac{1}{n} \sum_i \tilde{x}_{ig} \tilde{x}_{ik}| > 0.9$. The first $k \in \{10, 25, 50\}$ kept features are used, and k is a hyperparameter.

3.2.6 Classifiers

Four classic models are trained on each block, all with the sample weights w_i of Eq. (4) (implemented with scikit-learn [26] and LightGBM [27]).

(1) L2 logistic regression models the ADC probability with the sigmoid function and minimises the

weighted log-loss plus an L2 penalty:

$$p(x) = \frac{1}{1 + e^{-(\beta^\top x + \beta_0)}} \quad (18)$$

$$\min_{\beta, \beta_0} \frac{1}{2} \|\beta\|_2^2 + C \sum_{i=1}^n w_i [-y_i \log p(x_i) - (1 - y_i) \log(1 - p(x_i))] \quad (19)$$

A small C means a strong penalty, so the weights stay small and the model stays simple.

(2) Support vector machine [28] with labels $t_i = 2y_i - 1 \in \{-1, +1\}$ finds the boundary with the widest margin, allowing some violations ξ_i :

$$\min_{v, b, \xi} \frac{1}{2} \|v\|^2 + C \sum_i w_i \xi_i \text{ s.t. } t_i(v^\top \phi(x_i) + b) \geq 1 - \xi_i, \xi_i \geq 0 \quad (20)$$

Its decision function uses the radial basis function (RBF) kernel, which allows a curved boundary:

$$f(x) = \sum_i \alpha_i t_i K(x_i, x) + b, K(x, x') = \exp(-\gamma_K \|x - x'\|^2), \gamma_K = \frac{1}{p\text{Var}(X)} \quad (21)$$

(3) LightGBM [27] builds 200 small decision trees one after another. Each new tree h_m is fitted to the errors (negative gradients) of the current model and added with a learning rate $\eta = 0.03$, for $m = 1, \dots, 200$:

$$F_m(x) = F_{m-1}(x) + \eta h_m(x), r_{im} = y_i - \sigma(F_{m-1}(x_i)) \quad (22)$$

The trees have 4 or 8 leaves, at least 10 patients per leaf, and use 80% of patients and 50% of features per tree, which limits overfitting.

(4) Ridge logistic regression on all features is Eq. (19) with every feature of the block (no top-k) and a very small $C \in [10^{-4}, 10^{-2}]$. It captures weak signal that is spread thinly over many correlated features.

3.2.7 Ensemble and Score Fusion

Each model gives a real-valued score $f_m(x)$ (the decision function, or the probability for LightGBM). Because the four scores have different scales, each is standardised with the mean μ_m and standard deviation σ_m of its training scores and the four are averaged into a block score:

$$s_b(x) = \frac{1}{4} \sum_{m=1}^4 \frac{f_m(x) - \mu_m}{\sigma_m} \quad (23)$$

The two block scores are z-scored and averaged with equal, fixed weights, so nothing is fitted at this stage:

$$s(x) = \frac{1}{2} [z(s_{\text{shell}}(x)) + z(s_{\text{pos}}(x))], \hat{y} = \mathbb{1}[s(x) > 0] \quad (24)$$

A positive score means ADC. Since the training weights balance the classes, 0 is the natural threshold.

3.2.8 Evaluation Metrics

The primary metric is the site-balanced AUC of Eq. (5). For the unseen-hospital and external tests the ordinary AUC is used, since only one hospital is scored at a time. At the threshold of Eq. (24) we also report

$$\text{Se} = \frac{TP}{TP + FN}, \text{Sp} = \frac{TN}{TN + FP}, \text{BA} = \frac{\text{Se} + \text{Sp}}{2} \quad (25)$$

where ADC is the positive class. Uncertainty is given by a stratified bootstrap [29]: patients are

resampled with replacement inside each (hospital, class) group $B = 2000$ times, the metric is recomputed each time, and the 95% confidence interval is $[\hat{\theta}_{(2.5)}^*, \hat{\theta}_{(97.5)}^*]$. The amount of hospital information left in the features is measured by a *hospital detector*: a one-versus-rest logistic regression trained to predict the hospital, scored on the test fold,

$$\text{Leak} = \frac{1}{H} \sum_{h=1}^H \text{AUC}(\mathbb{1}[h_i = h], g_h(\tilde{x}_i)) \quad (26)$$

A value near 0.5 means the hospital can no longer be recognised. For visualisation we also use PCA, which projects the features onto the two eigenvectors u_1, u_2 of the covariance matrix $\Sigma = \frac{1}{n} \tilde{X}^T \tilde{X}$ with the largest eigenvalues.

3.3 System Block Diagram

Figure 4 shows the complete pipeline from a raw CT scan to the ADC/SCC decision. Every step is automatic, and the same frozen pipeline was applied to both external test sets. The four colours group the steps into image preparation, tumour and regions, features, and model.

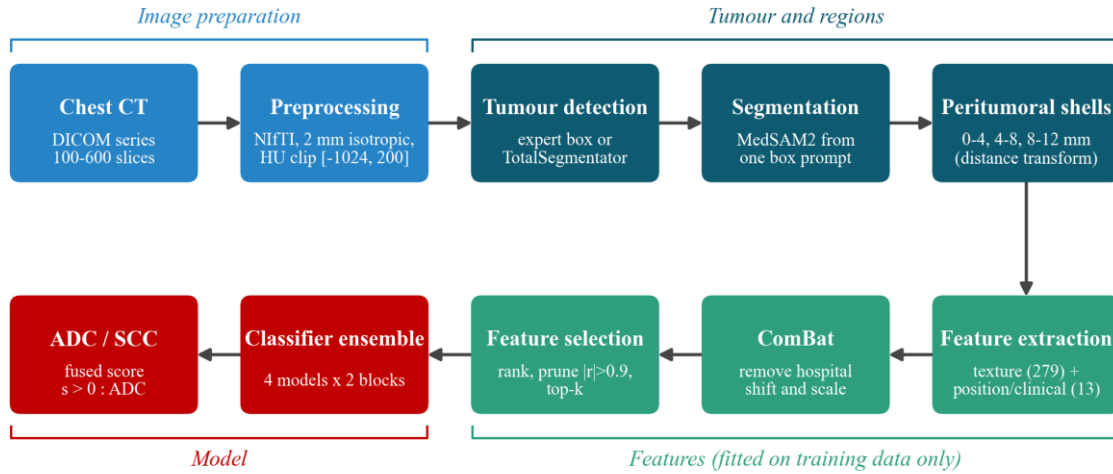


Figure 4: Block diagram of the complete pipeline. Steps in the lower row are fitted on training data only and then frozen.

- **Chest CT and preprocessing:** DICOM to NIfTI, 2 mm isotropic resampling, HU clipping (Eq. (7), Eq. (8)).
- **Tumour detection and segmentation:** expert box or TotalSegmentator box, then MedSAM2 outline.
- **Peritumoral shells:** three distance bands around the tumour (Eq. (10)).
- **Feature extraction:** 279 shell texture features and 13 position and clinical features.
- **ComBat, selection and classifiers:** Eq. (13) to Eq. (22), inside each training fold.
- **Decision:** fused score s of Eq. (24); $s > 0$ means ADC.

3.4 Model Architecture

Figure 5 shows the classifier part in detail. It has two parallel branches, one for each feature block. Each branch has its own preparation chain and its own four-model ensemble. The design is intentionally small: each model sees at most 50 features (or all 279 for the ridge model), so the whole classifier has a few hundred parameters plus the trees, compared with millions in a CNN. Table 3 lists

the settings.

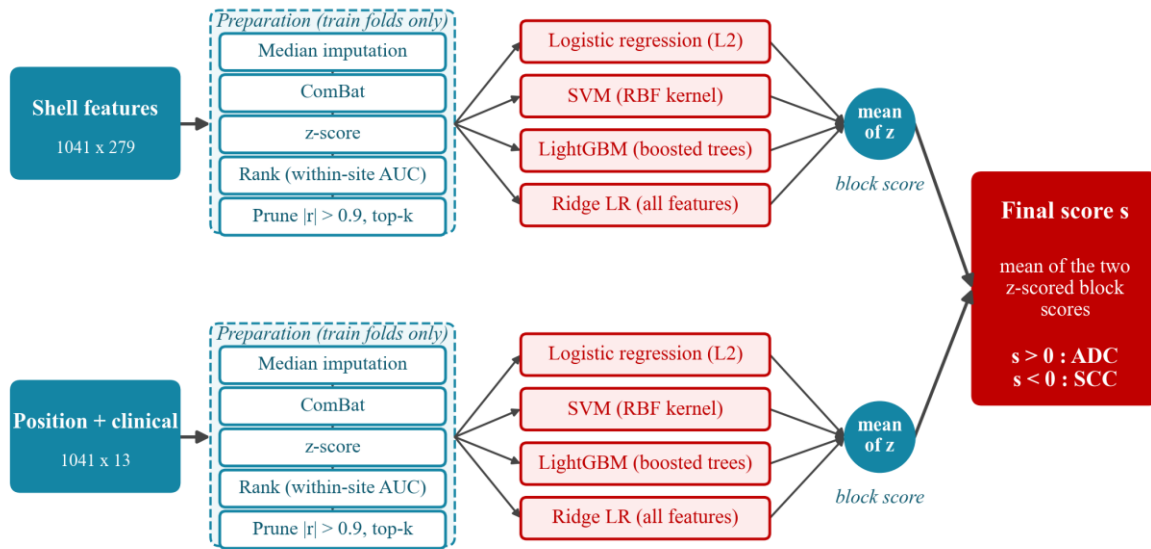


Figure 5: Architecture of the final classifier: two feature blocks, each with a preparation chain and a four-model ensemble, fused by averaging z-scored block scores.

Component	Setting
Input per patient	279 shell features + 13 position/clinical features
Missing values	drop feature if > 20% missing; else training median
Harmonisation	ComBat with empirical Bayes, per fold; label-free mapping of new hospitals
Selection	within-hospital AUC ranking, correlation pruning at 0.9, top-k with k in {10, 25, 50}
Logistic regression	L2, C in {0.01, 0.1, 1}
SVM	RBF kernel, gamma = scale, C in {0.1, 1, 10}
LightGBM	200 trees, learning rate 0.03, leaves in {4, 8}, min. 10 per leaf, subsample 0.8, feature fraction 0.5
Ridge (all features)	L2 logistic regression, C in {0.0001, 0.0003, 0.001, 0.003, 0.01}
Sample weights	$1 / (H \times n \text{ of the hospital-class group})$, Eq. (4)
Block score	mean of the four training-standardised scores, Eq. (23)
Final score	mean of the two z-scored block scores; ADC if $s > 0$, Eq. (24)

Table 3: Settings of the final model

3.5 Training and Validation Design

Two complementary validation schemes were used (Figure 6).

Repeated nested cross-validation. Patients are split into 5 folds, stratified by hospital and class so that every fold has all eight (hospital, class) groups. Four folds train the model and the fifth tests it. Inside the training part, an inner 3-fold cross-validation chooses the hyperparameters (k , C , number of leaves) by the site-balanced AUC; the model is then refitted on the whole training part. The outer loop is repeated with 5 random seeds (42-46), giving 25 test runs. Nested cross-validation avoids the optimistic bias that appears when hyperparameters are tuned on the test fold [30].

Leave-one-hospital-out (LOSO). The model is trained (with the same inner tuning) on three hospitals and tested on the fourth, which is mapped into the training reference by label-free ComBat. This is repeated for all four hospitals and is the closest simulation of using the model at a new hospital.

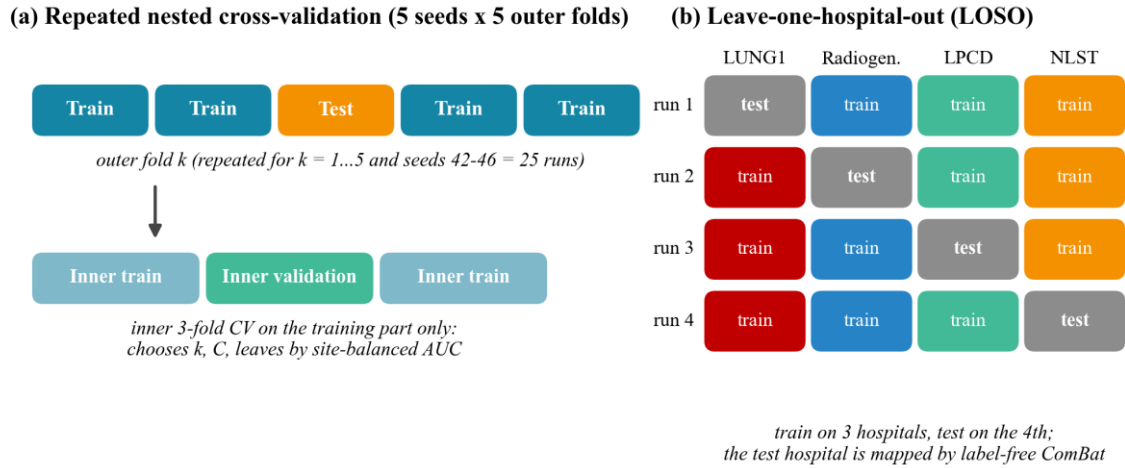


Figure 6: Validation design: (a) repeated nested cross-validation with an inner 3-fold loop for hyperparameter tuning; (b) leave-one-hospital-out testing.

Model	Hyperparameters searched	Most often chosen (25 folds)
Logistic regression	C in {0.01, 0.1, 1}; k in {10, 25, 50}	C = 0.01 (strongest penalty)
SVM (RBF)	C in {0.1, 1, 10}; k in {10, 25, 50}	C = 0.1
LightGBM	leaves in {4, 8}; k in {10, 25, 50}	4 leaves
Ridge (all)	C in {0.0001, 0.0003, 0.001, 0.003, 0.01}	C = 0.003 (shells), 0.01 (position)

Table 4: Hyperparameter grid; the selection metric is the site-balanced AUC of the inner folds

3.6 Locked External Test Protocol

A model that is tested, adjusted and tested again slowly turns its test set into a training set. To avoid this, the two external datasets were handled with the protocol of Figure 7. Before any external image was processed, a pre-registration document fixed the model (shells + position, frozen after Week 13), the exclusion rules and the primary metric. The images were processed by the frozen automatic pipeline. The model was fitted once on all 1041 development patients and the new dataset was mapped as one new site with label-free ComBat. The prediction file was saved together with its SHA-256 hash, a 64-character fingerprint that changes completely if even one byte of the file changes. Only then were the labels opened, once, by a scoring script that refuses to run a second time.

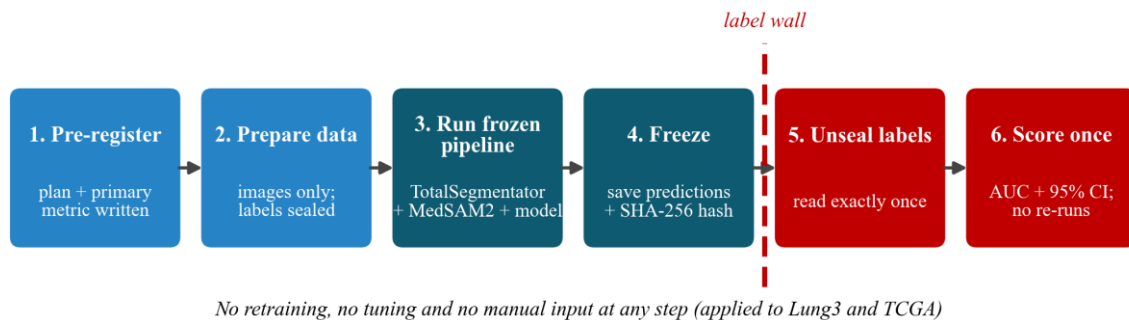


Figure 7: Protocol of the locked external tests. Labels are opened only after the predictions are frozen and hashed.

3.7 Core Source Code

The project was written in Python (about 90 scripts). The complete code with configuration files, saved splits and logs is in the project folder (see Section 7.3). The most important parts are listed below in the template's two-column code format: the site-balanced metric and feature ranking (Table 5), the ComBat harmoniser (Table 6), and nested model selection, ensembling and leave-one-hospital-out testing (Table 7).

<pre> def sb_weights(site, y): """1 / n_{site,label}, mean 1: every site counts equally, each site 50/50.""" key = pd.Series(site).astype(str) + " " + pd.Series(y).astype(str) cnt = key.map(key.value_counts()) w = 1.0 / cnt.values return w * len(w) / w.sum() def sb_auc(y, s, site): return float(roc_auc_score(y, s, sample_weight=sb_weights(site, y))) def univariate_score(X, y, site): """Mean over sites of the within-site AUC, folded around 0.5 (vectorised).""" sc = np.zeros(X.shape[1]) sites = np.unique(site) for g in sites: m = site == g yy = y[m] n1, n0 = yy.sum(), (1 - yy).sum() r = rankdata(X[m], axis=0) auc = (r[yy == 1].sum(0) - n1 * (n1 + 1) / 2.0) / (n1 * n0) sc += np.abs(auc - 0.5) return sc / len(sites) def site_leak_auc(Xtr, Xte, site_tr, site_te, y_tr, y_te): """mean one-vs-rest AUC of a hospital classifier (label-balanced weights).""" out = [] for s in np.unique(site_tr): t_tr, t_te = (site_tr == s).astype(int), (site_te == s).astype(int) if t_te.min() == t_te.max(): continue m = LogisticRegression(C=0.1, max_iter=5000).fit(Xtr, t_tr, sample_weight=sb_weights(site_tr, y_tr)) out.append(roc_auc_score(t_te, m.decision_function(Xte), sample_weight=sb_weights(site_te, y_te))) return float(np.mean(out)) </pre>	<pre> def select(self, k): if k == 0: # every feature (dense ridge model) return list(range(self.Xtr_.shape[1])) thr = self.cfg["features"][["correlation_prune"]] if not hasattr(self, "_sel_cache"): self._sel_cache = {} kmax = max(self.cfg["features"][["top_k_grid"]]) if "full" not in self._sel_cache: kept = [] C = None for j in self.order_: if len(kept) >= kmax: break if kept: v = self.Xtr_[j, :] r = np.abs(C.T @ v) / len(v) if r.max() > thr: continue kept.append(j) C = self.Xtr_[j, kept] self._sel_cache["full"] = kept return self._sel_cache["full"][:k] def fit_model(cfg, kind, hp, X, y, w, seed): if kind in ("logreg", "ridge_all"): mdl = LogisticRegression(C=hp["C"], max_iter=5000) mdl.fit(X, y, sample_weight=w) return mdl if kind == "svm": mdl = SVC(C=hp["C"], kernel="rbf", gamma=cfg["models"][["svm"]]["gamma"]) mdl.fit(X, y, sample_weight=w) return mdl import lightgbm as lgb g = cfg["models"][["lgbm"]] mdl = lgb.LGBMClassifier(n_estimators=g["n_estimators"], learning_rate=g["learning_rate"], num_leaves=hp["num_leaves"], min_child_samples=g["min_child_samples"], subsample=g["subsample"], subsample_freq=1, colsample_bytree=g["colsample_bytree"], random_state=seed, n_jobs=1, verbose=-1) mdl.fit(X, y, sample_weight=w) return mdl def score(mdl, X): if hasattr(mdl, "decision_function"): return mdl.decision_function(X) return mdl.predict_proba(X)[: , 1] </pre>
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Table 5: Source code: site-balanced weights and AUC, within-hospital feature ranking, hospital detector (left); correlation-pruned top-k selection and the four classifiers (right)

<pre> def _eb_prior(gamma_hat, delta_hat): g_bar, t2 = gamma_hat.mean(), gamma_hat.var(ddof=1) m, v = delta_hat.mean(), delta_hat.var(ddof=1) v = max(v, 1e-12) a = (2.0 * v + m * m) / v b = (m * v + m ** 3) / v return g_bar, max(t2, 1e-12), a, b def _eb_shrink(gamma_hat, delta_hat, n, iters=100, tol=1e-4): g_bar, t2, a, b = _eb_prior(gamma_hat, delta_hat) g, d = gamma_hat.copy(), delta_hat.copy() for _ in range(iters): g_new = (n * t2 * gamma_hat + d * g_bar) / (n * t2 + d) ss = n * (delta_hat + (gamma_hat - g_new) ** 2) d_new = (0.5 * ss + b) / (n / 2.0 + a - 1.0) conv = max(np.max(np.abs(g_new - g) / (np.abs(g) + 1e-8)), np.max(np.abs(d_new - d) / (np.abs(d) + 1e-8))) g, d = g_new, d_new if conv < tol: break return g, d </pre>	<pre> class Harmoniser: def __init__(self, method="none", empirical_bayes=True): self.method = method self.eb = empirical_bayes def fit(self, X, site, y): X = np.asarray(X, float) site = np.asarray(site) self.sites_ = list(np.unique(site)) if self.method == "none": return self bal = self.method == "balanced_combat" loc, var, n = {}, {}, {} for s in self.sites_: m = site == s loc[s], var[s] = _site_stats(X[m], None if y is None else np.asarray(y)[m], bal) n[s] = int(m.sum()) if bal: # every site counts equally in the reference self.alpha_ = np.mean([loc[s] for s in self.sites_], axis=0) pooled = np.mean([var[s] for s in self.sites_], axis=0) else: tot = sum(n.values()) self.alpha_ = sum(n[s] * loc[s] for s in self.sites_) / tot pooled = sum(n[s] * var[s] for s in self.sites_) / tot self.sigma_ = np.sqrt(np.maximum(pooled, 1e- 12)) self.gamma_, self.delta_ = {}, {} for s in self.sites_: gh = (loc[s] - self.alpha_) / self.sigma_ dh = np.maximum(var[s] / self.sigma_ ** 2, 1e-8) if self.eb and len(gh) > 2: gh, dh = _eb_shrink(gh, dh, n[s]) self.gamma_[s], self.delta_[s] = gh, dh return self def transform(self, X, site): X = np.asarray(X, float) if self.method == "none": return X.copy() site = np.asarray(site) out = np.empty_like(X) for s in np.unique(site): if s not in self.gamma_: raise KeyError("site %r unseen at fit time; use fit_new_site" % s) m = site == s z = (X[m] - self.alpha_) / self.sigma_ out[m] = self.sigma_ * (z - self.gamma_[s]) / np.sqrt(self.delta_[s]) + self.alpha_ return out def fit_new_site(self, X_new, name, balanced, w1=None): X_new = np.asarray(X_new, float) loc, var = _site_stats(X_new, None, balanced, w1) self.gamma_[name] = (loc - self.alpha_) / self.sigma_ self.delta_[name] = np.maximum(var / self.sigma_ ** 2, 1e-8) return self </pre>
--	--

Table 6: Source code: empirical-Bayes priors and shrinkage (left); ComBat harmoniser with label-free mapping of a new hospital (right)

<pre> def select_and_fit(cfg, X, y, site, seed, harm): """Inner-CV hyper-parameter choice on (X, y, site), then refit on all of it.""" strata = np.array([f"{a} {b}" for a, b in zip(site, y)]) inner = StratifiedKFold(cfg["experiment"]["n_inner_folds"], shuffle=True, random_state=seed) sc = {k: {i: [] for i in range(len(grid(cfg, k)))} for k in MODELS} for a, b in inner.split(X, strata): pr = Prep(harm, cfg).fit(X[a], y[a], site[a]) Xa, Xb = pr.Xtr_, pr.transform(X[b], site[b]) w = sb_weights(site[a], y[a]) for k in MODELS: for i, hp in enumerate(grid(cfg, k)): sel = pr.select(hp["k"]) m = fit_model(cfg, k, hp, Xa[:, sel], y[a], w, seed) sc[k][i].append(sb_auc(y[b], score(m, Xb[:, sel]), site[b])) pr = Prep(harm, cfg).fit(X, y, site) w = sb_weights(site, y) fitted = {} for k in MODELS: best = max(sc[k], key=lambda i: np.mean(sc[k][i])) hp = grid(cfg, k)[best] sel = pr.select(hp["k"]) m = fit_model(cfg, k, hp, pr.Xtr[:, sel], y, w, seed) s = score(m, pr.Xtr[:, sel]) fitted[k] = (m, sel, s.mean(), s.std() + 1e-8, dict(hp, inner=float(np.mean(sc[k][best])))) return pr, fitted def ens(fitted, Z): return np.mean([(score(m, Z[:, sel]) - mu) / sd for m, sel, mu, sd, _ in fitted.values()], axis=0) </pre>	<pre> def loso_one(cfg, X, y, site, held, seed, harm): tr, te = site != held, site == held pr, fitted = select_and_fit(cfg, X[tr], y[tr], site[tr], seed, harm) # label-free mapping of the held-out hospital into the training reference Xte_raw = X[te][:, pr.keep_] Xte_raw = np.where(np.isnan(Xte_raw), pr.med_, Xte_raw)[: , pr.keep2_] if harm != "none": pr.h_.fit_new_site(Xte_raw, held, balanced=False) Zte = (pr.h_.transform(Xte_raw, np.array([held] * te.sum())) - pr.mu_) / pr.sd_ s = ens(fitted, Zte) return pd.DataFrame({"idx": np.where(te)[0], "seed": seed, "held_out": held, "score": s}) </pre>
--	--

Table 7: Source code: inner-CV hyperparameter selection and ensemble (left); leave-one-hospital-out test with label-free harmonisation (right)

4 Implementation

4.1 Description

Environment. All work ran on one laptop. Table 8 lists the hardware and main software. Three separate Python environments were needed because MedSAM2, TotalSegmentator and the main pipeline require different PyTorch versions.

Item	Details
Hardware	Laptop, NVIDIA RTX 3050 GPU (4 GB), 16 GB RAM, Windows 11
Language	Python 3.9 (main pipeline), 3.12 (MedSAM2)
Image I/O and resampling	SimpleITK, pydicom, dcm2niix-style conversion, NumPy, SciPy
Features	PyRadiomics 3 (texture), own code for position features and shells
Machine learning	scikit-learn 1.5.2, LightGBM 4.6.0, PyTorch 2.6 (deep baselines), MONAI 1.3.2
Segmentation	MedSAM2 (box-prompted), TotalSegmentator 2.18 (lung nodules)
Data download	NBIA Data Retriever (TCIA), s5cmd and idc-index (IDC)
Reporting	Matplotlib, python-pptx, python-docx

Table 8: Implementation environment

Data preparation. The DICOM series were downloaded (about 90 GB in total), checked for geometry (spacing, orientation, missing slices) and converted to NIfTI. Figure 8 shows one LUNG1 scan before and after preprocessing. In the original grid the coronal view looks squashed because the slices are 3 mm apart while pixels are about 1 mm; after resampling the anatomy has the correct proportions and the HU window makes lung and soft tissue easy to see.

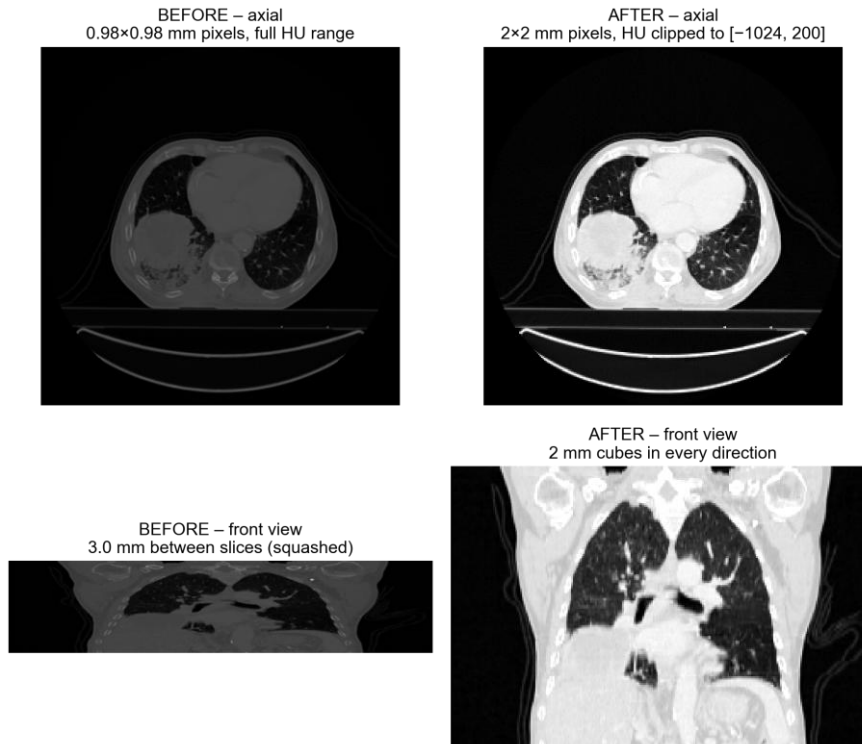


Figure 8: Preprocessing of patient LUNG1-001: original axial and coronal views (left) and the same views after 2 mm isotropic resampling and HU clipping (right).

Segmentation. Our first segmentation method kept voxels above -750 HU inside the box and cleaned the result with morphology. It worked with tight boxes but leaked into the chest wall when the box was loose, which was the case for the NLST boxes. MedSAM2 was therefore adopted for all 1041 patients (38 minutes on the laptop GPU, no failures). The settings were chosen on 40 expert outlines using the worst-case Dice, without looking at any ADC/SCC label: lung window [-1350, 150] HU, full 512 x 512 slices, one box prompt on the slice with the largest box, propagation in both directions, and the result limited to the box enlarged by 3 mm. Figure 9 shows an example and Figure 10 the validation on all 343 expert outlines.

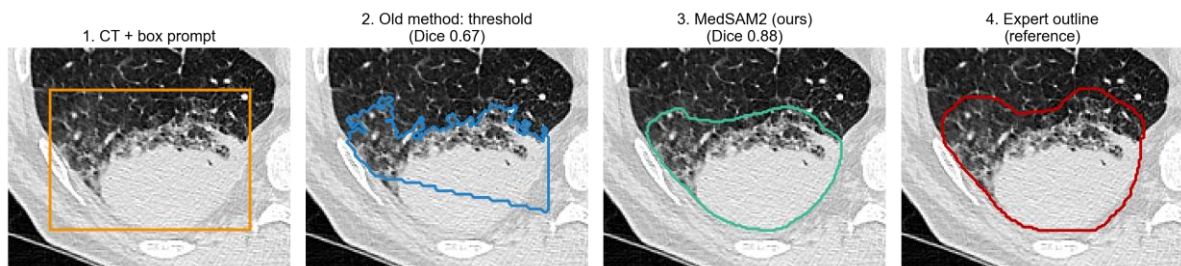


Figure 9: Segmentation of one Radiogenomics patient: box prompt, old threshold method (Dice 0.67), MedSAM2 (Dice 0.88) and the expert outline.

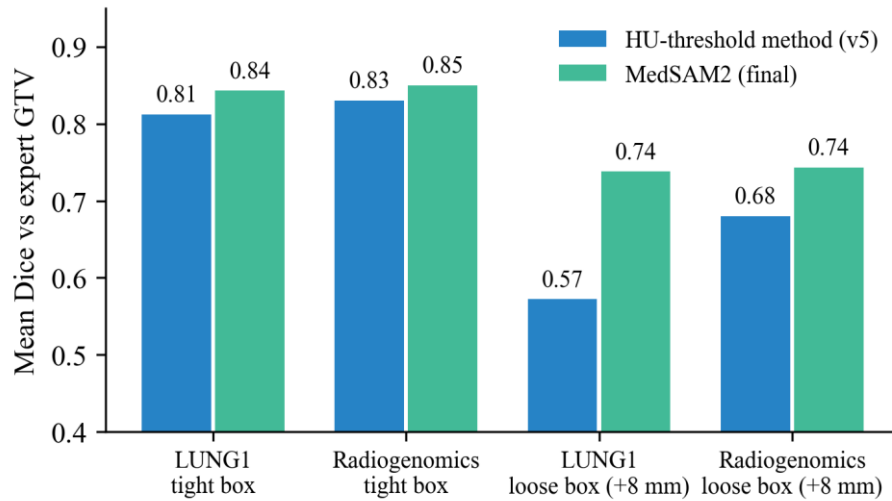


Figure 10: Mean Dice against expert outlines for tight boxes and loose boxes (+8 mm on each side, which imitates an automatic detector). MedSAM2 is better in all four cases and much better with loose boxes.

Feature extraction. Shells were built with a Euclidean distance transform (Eq. (10)); Figure 11 shows them on a real scan. PyRadiomics extraction for all regions took about 20 minutes and position features about 13 minutes using parallel jobs, with no failures in the development data. Extraction was run once and the features were saved, so model training never recomputed them.

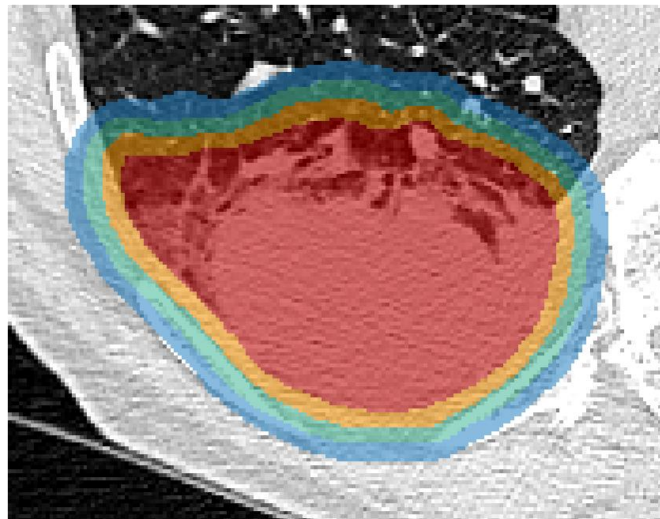


Figure 11: Tumour (GTV) and the three peritumoral shells at 0-4, 4-8 and 8-12 mm on an axial CT slice.

Training. One full 5 x 5 nested cross-validation of one feature block takes about 10 seconds to a few minutes with 14 parallel jobs, because the models are small. All splits, random seeds, chosen hyperparameters, out-of-fold predictions, metrics and package versions were saved for reproducibility.

4.2 Implementation Challenges

- **Different DICOM types.** Eight Lung-PET-CT-Dx patients had annotations only on a secondary-capture image without 3-D geometry; they were excluded by rule.
- **Loose boxes in NLST.** The NLST boxes were about three times larger than the tumour, so the threshold segmentation absorbed chest wall. This was solved by MedSAM2.
- **Software conflicts.** Installing one foundation-model package replaced the GPU version of PyTorch with a CPU version during a run. The environment was repaired and later installs used --no-deps.

- **Large downloads on Windows.** The s5cmd tool failed to rename files across drives, so downloads were run from the destination drive with relative paths.
- **Limited GPU memory (4 GB).** Fine-tuning used a frozen backbone with only the last block trained, and cached augmented crops.

5 Design Analysis and Evaluation

5.1 Novelty

We did not invent ComBat, MedSAM2 or weighted AUC individually. The novelty of the project lies in how they are combined and in how the model is evaluated:

- **Explicit measurement of the hospital shortcut for ADC/SCC.** We show that the hospital name alone gives a pooled AUC of 0.71 and use a site-balanced AUC that provably gives 0.5 to any hospital-only predictor (Eq. (6)). The honest score improved from 0.642 (naive model) to 0.722.
- **Fully automatic, uniform tumour outlining.** TotalSegmentator plus a single MedSAM2 box prompt gives masks at Dice 0.85 against experts, with the same method at every hospital so that the mask source is not a hidden fingerprint.
- **Four hospitals, 1041 patients, three countries.** To our knowledge this is one of the largest public multi-hospital sets used for this task, and it allowed leave-one-hospital-out testing (0.714).
- **Locked, pre-registered external tests.** The plan was written first, predictions were hashed before the labels were opened, and each test was scored once (Lung3 0.686, TCGA 0.585).
- **A fair comparison of simple and deep models.** Attention-MIL, CNN features, two foundation models and adversarial fine-tuning were all tested under the same protocol and all scored lower.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to public health and safety

The model is designed as decision support. Its output is a score, not a diagnosis, and a biopsy is still needed for treatment decisions and gene testing. With a sensitivity of 0.61 and a specificity of 0.68 at the default threshold (Section 5.3.3), a wrong prediction is common enough that the score must never be used alone. Several safety choices follow from this: the threshold is fixed in advance, the model reports results per hospital, and we state clearly that it should not be used on 5 mm CT, where external performance was at chance. Because the pipeline uses only the CT that the patient already has, it adds no radiation dose and no procedural risk.

5.2.2 Considerations to environment

We chose small classical models instead of large networks. Training the final model takes minutes on a laptop GPU, and the whole project used roughly 10-15 kWh of electricity (Section 8). Public data was reused instead of collecting new scans, which avoids extra scanning and its energy and radiation cost. Downloaded data was kept on one disk and processed once, and extracted features were stored so that experiments did not repeat expensive computation.

5.2.3 Considerations to cultural and societal needs

In Bangladesh and similar countries, biopsy facilities and pathologists are concentrated in large hospitals, while CT scanners are more widely available. A CT-based tool could help prioritise patients for biopsy or guide the choice of sample. Smoking rates are high among men, which raises the share of SCC. Our data come from the Netherlands, the USA and China, so performance in a South Asian population is unknown and must be tested locally before any use. The site-balanced evaluation is also a societal consideration: it makes sure that a hospital with fewer patients is not ignored by the metric.

5.3 Investigations (PO(d))

5.3.1 Design of Experiment

Each experiment changed one part of the system while the folds, seeds and metric stayed the same, so that differences could be attributed to that part. Table 9 lists the experiments.

#	Question	What was varied	Metric
E1	How strong is the hospital shortcut?	hospital-only predictor vs naive vs final model	pooled and site-balanced AUC
E2	Does attention over local patches help?	mean vs gated-attention MIL, patch features, ROI	AUC (LUNG1)
E3	Which feature block carries signal?	7 blocks alone and 127 fusions	site-balanced AUC, LOSO
E4	Does harmonisation remove the fingerprint?	ComBat on/off	hospital-detector AUC, AUC
E5	Do better masks help?	threshold masks vs MedSAM2	Dice, AUC
E6	Do more hospitals help?	2, 3 and 4 training hospitals	LOSO AUC
E7	Do deep features beat radiomics?	FMCIB, CT-FM, fine-tuned CNN with adversary	AUC, hospital-detector AUC
E8	Does the frozen model transfer?	locked Lung3 and TCGA	AUC with 95% CI

Table 9: Design of experiments

5.3.2 Data Collection

All data are public and de-identified. Table 10 summarises the six cohorts. The development set has 1041 patients (697 ADC, 344 SCC) from four sources [3, 31, 32, 33]. NLST tumour boxes come from the Sybil project [34]. Lung3 [35] and TCGA [36] were reserved as external test sets and were not opened until the end.

Cohort	Country / setting	Patients (ADC/SCC)	Typical slice	Tumour location	Role
LUNG1 (NSCLC-Radiomics)	Netherlands, radiotherapy, mostly stage III	202 (51/151)	3 mm	expert outline	development
NSCLC-Radiogenomics	USA, surgical	141 (112/29)	1.25 mm	expert outline	development
Lung-PET-CT-Dx	China, diagnostic	301 (244/57)	mostly 5 mm	expert boxes	development
NLST	USA, screening trial	397 (290/107)	2 mm	expert boxes (Sybil)	development
Lung3 (Radiomics-Genomics)	Netherlands, surgical	79 (44/35)	4-5 mm	automatic	locked test
TCGA-LUAD / LUSC	USA, 9 hospitals	84 (48/36)	5 mm (2/3 of scans)	automatic	locked test

Table 10: Datasets used in this project

Inclusion rules. Patients needed a pre-treatment CT series and a pathology label of ADC or SCC. Mixed or unclear types were excluded: in NLST 155 patients whose first cancer was not ADC or SCC

and 29 with more than one lung primary; in Lung-PET-CT-Dx 8 patients with secondary-capture images; in Lung3 10 patients with NSCLC not otherwise specified, large cell neuroendocrine or mixed histology; in TCGA 19 patients without a usable CT and 1 with no detected nodule. The selection used only metadata, never image-based predictions.

5.3.3 Results and Analysis

(a) Early approach: attention-MIL on local patches. The project began with a weakly supervised attention-MIL model on LUNG1 [19, 20], where each patient was a bag of 1 cm radiomic patches. Table 11 summarises these trials, which were shown at the progress presentation. Attention did not reliably beat simple mean pooling, and the best result after adding a second cohort (0.675) was later found to be partly inflated by the hospital shortcut, since a hospital detector reached AUC 0.998 on those two cohorts. This led to the redesign described in this report.

Trial	Setup	Result (AUC)	Finding
1	Gradient ROI, 29,184 patches, 74 features	0.604 / 0.568	attention unstable, lost to mean pooling
2	GTV + 8 mm rim, 37,464 patches	0.570 mean, 0.577 attention	wider region did not clearly help
3	mRMR [37] / LASSO [38] selection inside CV	0.593	small gain, CI included zero
4	Frozen ResNet-18 [39] patch embeddings	0.549 mean, 0.615 attention	first clear attention gain
5	CLAM-style instance loss	0.615 to 0.568	sharper attention, lower AUC
6	Global GTV and rim radiomics, late fusion	0.567 / 0.537	global models also weak on LUNG1
7	LUNG1 + Radiogenomics merged	0.675 (pooled)	partly hospital shortcut

Table 11: Summary of the early attention-MIL phase (before the progress presentation)

(b) The hospital shortcut (E1). As shown in Figure 2, the hospital name alone gives a pooled AUC of 0.710 and the naive radiomics model 0.727. Under the site-balanced AUC these become 0.497 and 0.642. All later results use the site-balanced metric.

(c) Main cross-validation result. The final model reached a site-balanced AUC of **0.722 (95% CI 0.687-0.762)** over 5 seeds x 5 folds on 1041 patients, with a standard deviation over seeds below 0.01. Figure 12 shows the ROC curve inside each hospital, built from out-of-fold predictions averaged over the seeds. Every curve is above the diagonal. Lung-PET-CT-Dx is the easiest hospital (0.85) and LUNG1 the hardest (0.63). In LUNG1, 78% of patients are stage III, and at that stage ADC and SCC tumours have similar sizes (median about 44 versus 48 ml), so size and position carry little information.

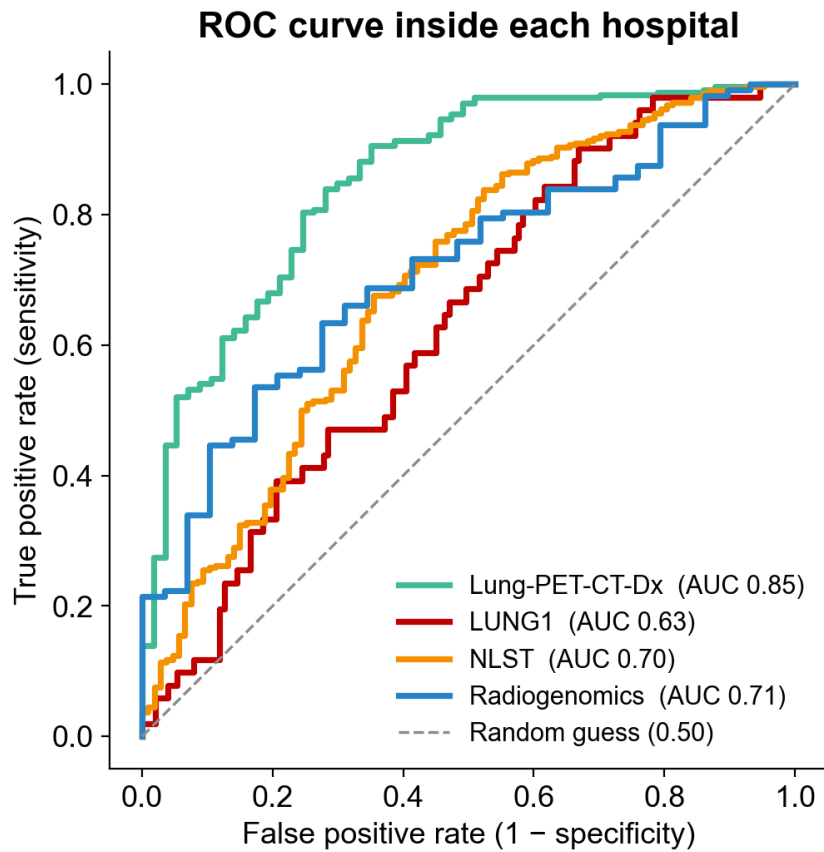


Figure 12: ROC curves of the final model inside each hospital (out-of-fold predictions, 5 x 5 cross-validation). The dashed line is random guessing.

Figure 13 shows the distribution of the fused score by hospital and class. In every hospital the ADC box lies above the SCC box, but the overlap is large, which matches an AUC around 0.7. Table 12 and Figure 14 give the results at the fixed threshold $s = 0$.

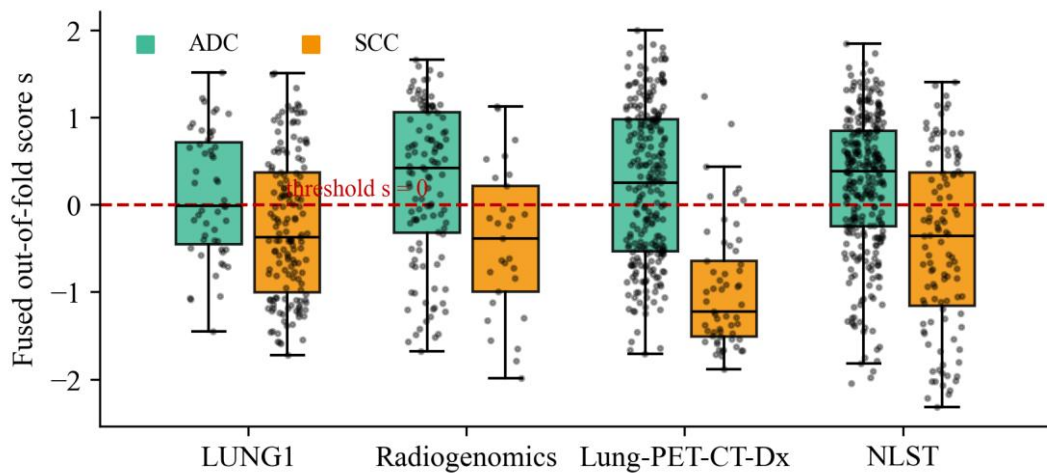


Figure 13: Fused out-of-fold score by hospital and class. Points are patients; the red line is the decision threshold.

Hospital	Patients (ADC/SCC)	AUC	Sensitivity	Specificity	Balanced acc.
LUNG1	202 (51/151)	0.631	0.490	0.623	0.556
Radiogenomics	141 (112/29)	0.710	0.616	0.724	0.670
Lung-PET-CT-Dx	301 (244/57)	0.852	0.561	0.877	0.719
NLST	397 (290/107)	0.694	0.662	0.645	0.653

Hospital	Patients (ADC/SCC)	AUC	Sensitivity	Specificity	Balanced acc.
All (site-balanced)	1041 (697/344)	0.722	0.607	0.680	0.644

Table 12: Cross-validation results per hospital at the threshold $s = 0$

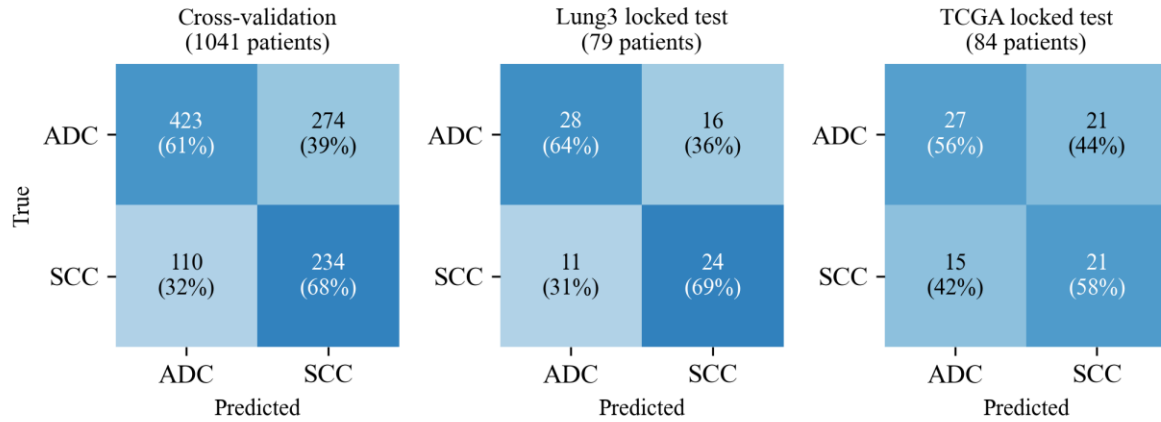


Figure 14: Confusion matrices at the threshold $s = 0$ for cross-validation and the two locked external tests. Percentages are per true class.

(d) Unseen hospitals (LOSO, E6). Figure 15 compares each hospital's score when it was part of the training data (pooled CV) with its score when it was left out completely. The drop is very small (at most 0.017), and the LOSO mean is **0.714**, almost equal to the pooled 0.722. This suggests that the model does not rely on memorised hospital style. More hospitals helped: with three training hospitals the LOSO mean was 0.678, and adding NLST raised it to 0.704 with the earlier masks, mostly by improving the held-out Lung-PET-CT-Dx score from 0.735 to 0.839.

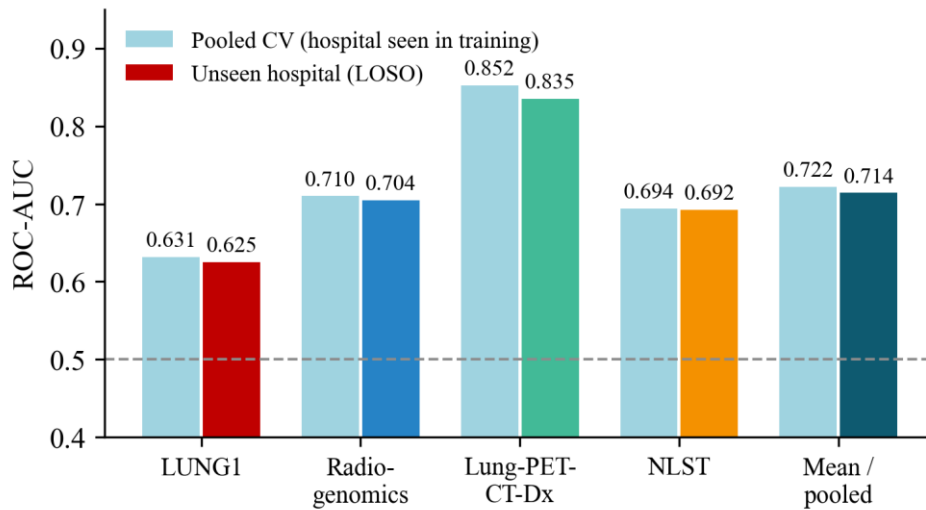


Figure 15: AUC of each hospital in pooled cross-validation (light bars) and when the hospital was left out of training (coloured bars).

(e) Harmonisation (E4). Figure 16 shows the shell features projected by PCA. Before ComBat, LUNG1 forms its own cloud; after ComBat the hospitals overlap. Figure 17 measures this with the hospital detector of Eq. (26): it fell from 0.81-0.91 to 0.56-0.64 for the radiomics and position blocks. The site-balanced AUC changed by only -0.004 to +0.020 per block. ComBat therefore does not raise the score; it makes the score more trustworthy, because the remaining performance cannot come from scanner style.

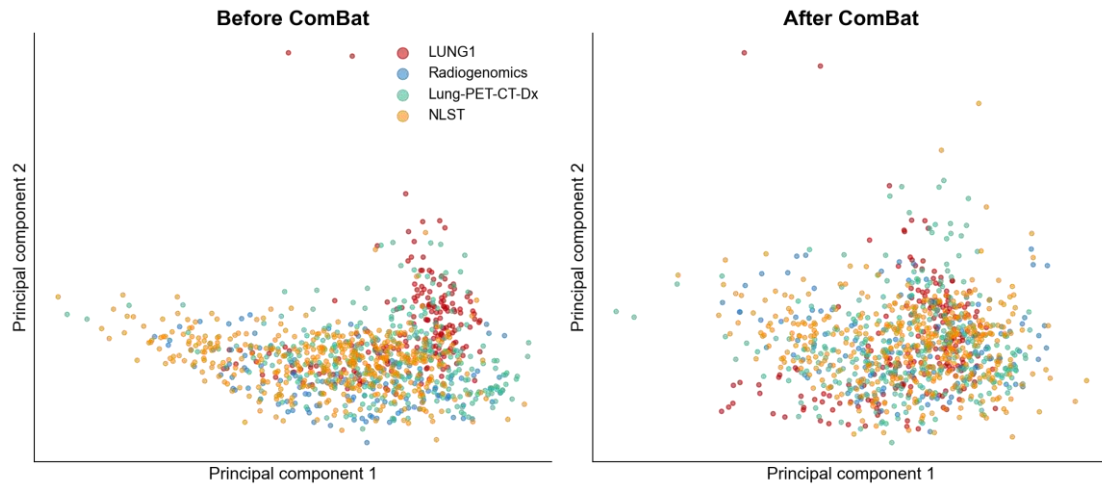


Figure 16: PCA of the 279 shell features before and after ComBat; each point is a patient, coloured by hospital.

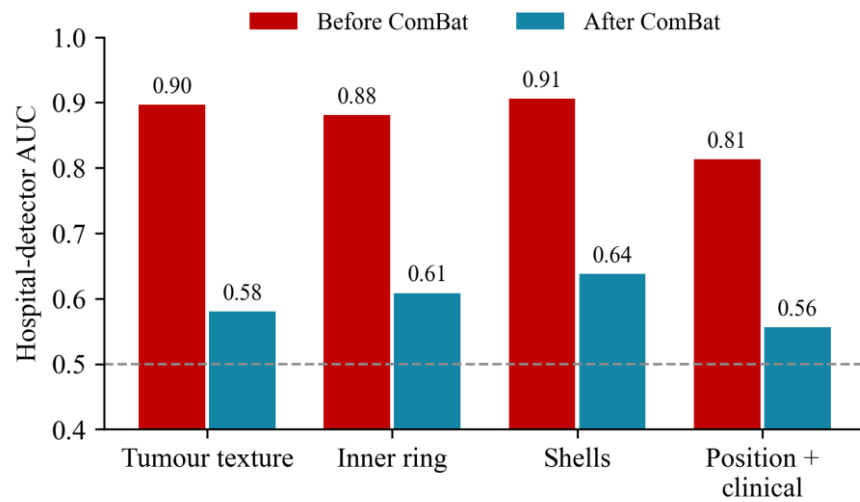


Figure 17: Hospital-detector AUC before and after ComBat (three-hospital stage). 0.5 means the hospital cannot be recognised.

(f) Ablation (E3, E5). Figure 18 shows the site-balanced AUC of each feature block and of the fusions, all with MedSAM2 masks and ComBat. The handcrafted blocks are close to each other (0.698-0.709), the two deep-feature blocks are clearly lower, and fusing shells with position gives the best pre-specified result. Table 13 lists the effect of each design change.

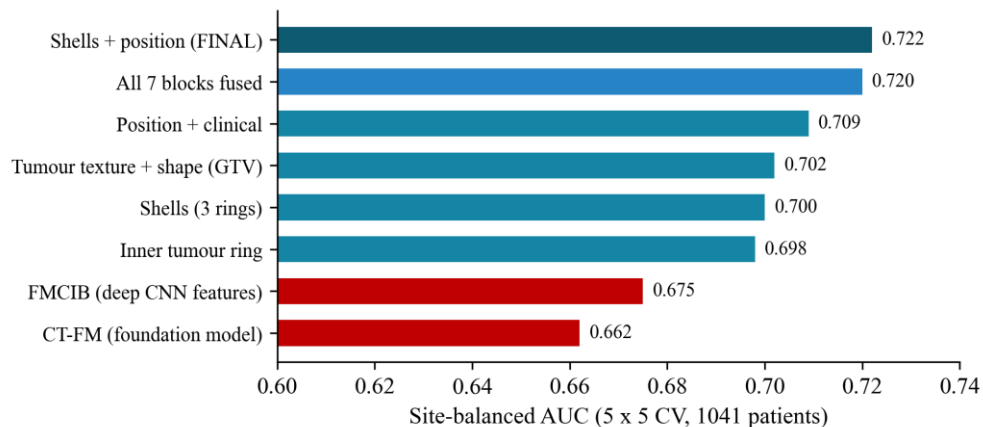


Figure 18: Ablation: site-balanced AUC of single feature blocks and fusions (5 x 5 CV, 1041 patients). Red bars are deep features. Note that the x-axis starts at 0.6.

Change	Before	After	Conclusion
Threshold masks to MedSAM2 masks	0.714	0.722	better masks give better features
3 to 4 training hospitals (LOSO mean)	0.682	0.704	more diverse data helps transfer
Best single block to shells + position	0.709	0.722	the two blocks are complementary
Shells + position to all 7 blocks	0.722	0.720	more features add noise, not signal
ComBat off to on (hospital detector)	0.81-0.91	0.56-0.64	fingerprint largely removed
Single model to 4-model ensemble (shells)	0.677-0.697	0.700	ensemble is more stable

Table 13: Effect of individual design changes (site-balanced AUC unless stated)

(g) Ensemble and tuning. Figure 19 compares each classifier with the ensemble. No single model is best in both blocks (ridge is best for shells, SVM for position), but the four-model ensemble is better than every single model in both blocks, and fusing the two blocks adds a further 0.013. Figure 20 shows which hyperparameters the inner cross-validation chose in the 25 outer folds. The simplest option won almost every time: $C = 0.01$ for logistic regression, $C = 0.1$ for the SVM and 4-leaf trees. This is a sign that the data are noisy and small, and that more flexible models would mainly fit noise.

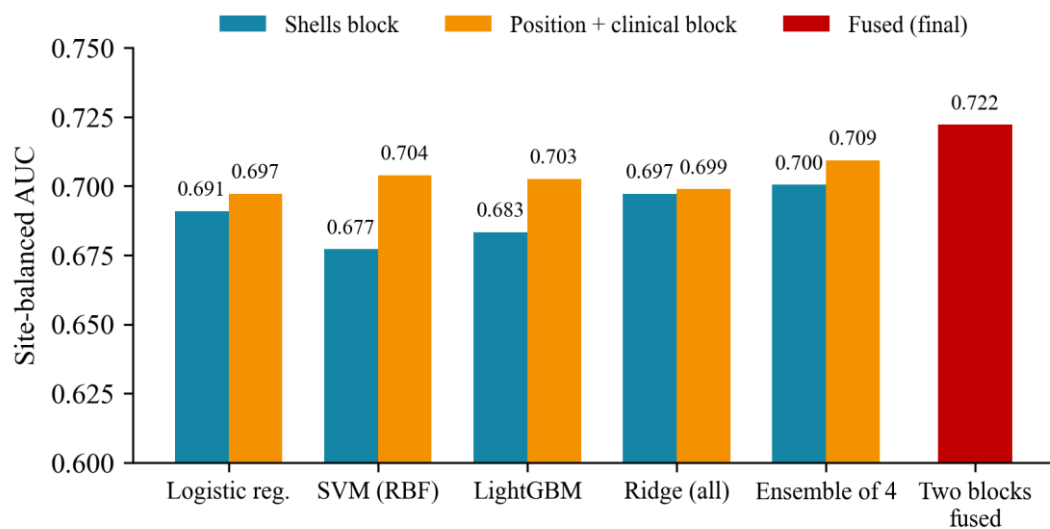


Figure 19: Site-balanced AUC of each classifier, of the four-model ensemble per block, and of the fused final model.

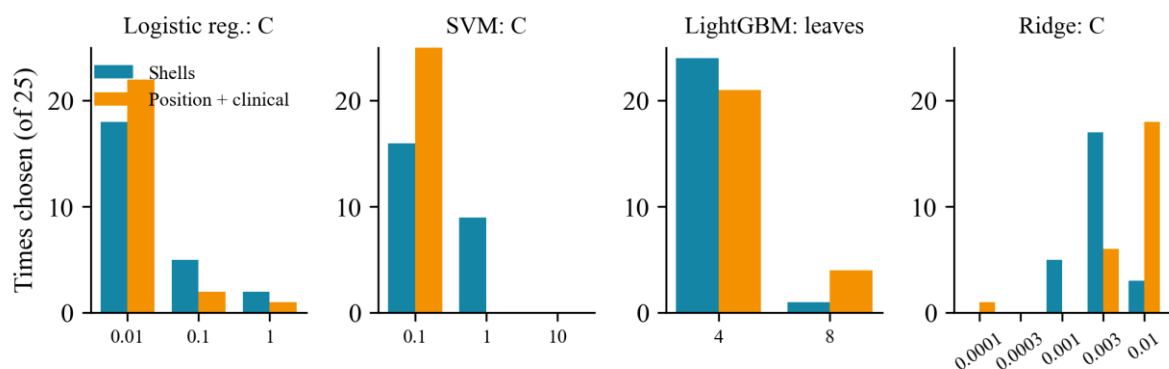


Figure 20: How often each hyperparameter value was chosen by the inner 3-fold cross-validation in the 25 outer folds.

(h) Deep-learning alternatives (E7). Table 14 lists the larger models that were tried under the same protocol. Frozen FMCIB features [15] reached 0.675. CT-FM [16] reached 0.662 and kept hospital information even after ComBat (detector 0.765), probably because its whole-CT pre-training encodes scanner style. Fine-tuning the last block of FMCIB (35 million parameters) with a gradient-reversal hospital adversary [40] overfitted within a few epochs and scored 0.606, with a detector AUC of 0.815.

With about 660 training patients per fold, the networks learned hospital and noise faster than histology.

Approach	Site-balanced AUC	LOSO mean	Hospital detector	Finding
Attention-MIL on FMCIB crop bags	0.669	0.634	n/a	did not beat mean pooling
FMCIB, frozen (3-D ResNet-50)	0.675	0.660	0.263	weaker than handcrafted features
CT-FM, frozen	0.662	0.652	0.765	kept hospital identity
Fine-tuned CNN	0.615	0.644	0.814	overfitted
Fine-tuned CNN + adversary	0.606	0.651	0.815	adversary did not remove hospital
Final model (shells + position)	0.722	0.714	0.54-0.61	best and simplest

Table 14: Deep-learning alternatives compared with the final model

(i) Project journey. Figure 21 shows the main AUC at each stage of the project. The points use different data and metrics, so the plot tells the story rather than giving a strict comparison. The largest improvements after the progress presentation came from more hospitals, a fair metric and better masks, not from bigger models.

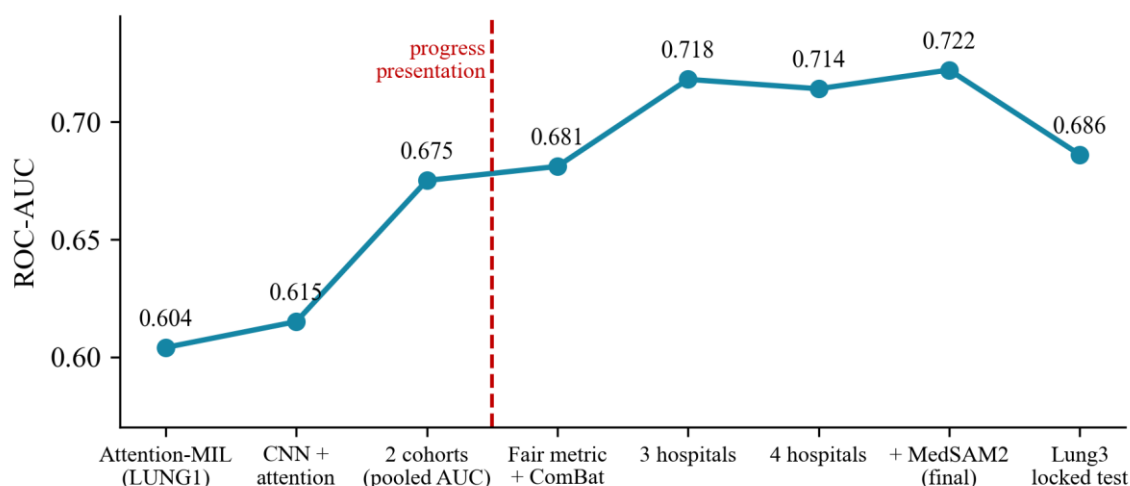


Figure 21: AUC at each stage of the project. The red line marks the progress presentation.

(j) Locked external tests (E8). Figure 22 and Table 15 show the two pre-registered results. On **Lung3** the model reached **0.686** (95% CI 0.557-0.807), close to the expected LOSO level of 0.714. On **TCGA** it reached **0.585** (0.462-0.707). Slice thickness was important in both tests: on thin slices (at most 2.5 mm) the AUC was 0.90 on Lung3 and 0.73 on TCGA, while on thick slices it was 0.68 and 0.51. These subgroups are small (18 and 29 thin-slice patients), so they are hints rather than proof. TCGA has a further problem: each of its 9 contributing hospitals sent only ADC or only SCC patients, so in TCGA the hospital is the label. A model designed to ignore hospital style cannot use this, and the effective sample size is closer to 9 hospitals than to 84 patients. TCGA tumours were also smaller (median 16 versus 29 ml) and were scanned with more reconstruction kernels (19 versus 10).

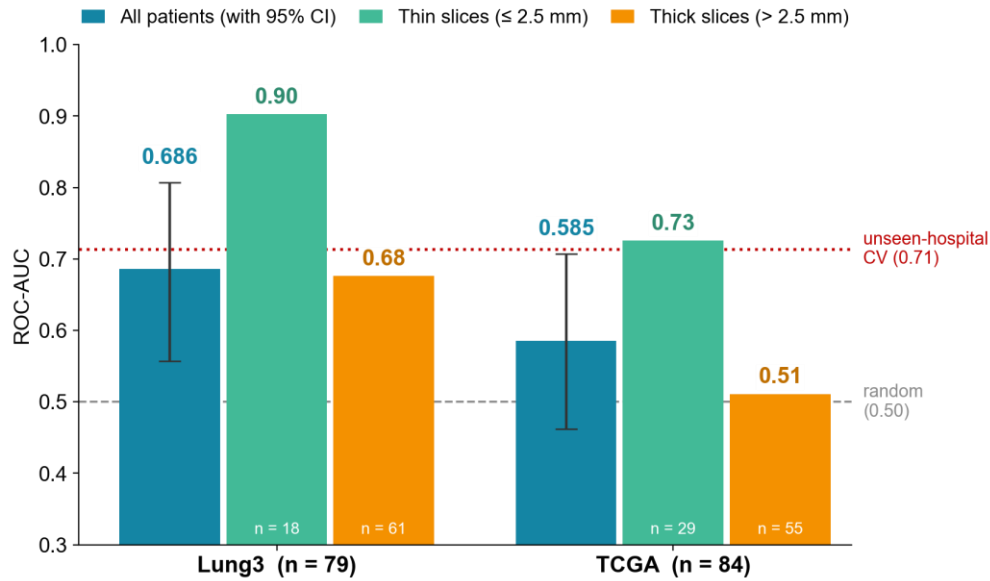


Figure 22: Locked external tests: AUC on all patients with 95% bootstrap CI, and on thin (at most 2.5 mm) and thick slices. The red dotted line is the LOSO level (0.71).

Test set	n (ADC/SCC)	AUC [95% CI]	Se	Sp	BA	Thin slices	Thick slices
Lung3	79 (44/35)	0.686 [0.557, 0.807]	0.636	0.686	0.661	0.903 (n=18)	0.676 (n=61)
TCGA	84 (48/36)	0.585 [0.462, 0.707]	0.562	0.583	0.573	0.726 (n=29)	0.511 (n=55)

Table 15: Locked external test results of the pre-registered model. The thin/thick split was pre-declared for Lung3 and post hoc for TCGA.

(k) Comparison with existing work. Table 16 places our results next to published studies. Scores above 0.8 mostly come from one hospital or from random splits of pooled data. On truly independent hospitals the published range is about 0.54-0.73. On the same external dataset (Lung3) our locked result of 0.686 is higher than the 0.60 of Chaunzwa *et al.*, while Song *et al.* report 0.80 on Lung3 and 0.82 on TCGA. Their numbers were obtained with expert tumour masks, by choosing the best of 130 models on the test sets, and with hospitals mixed randomly in training, so the two results were produced under different rules.

Study	Data	How it was tested	AUC	Remarks
Zhu 2018 [9]	129 pts, 1 hospital	split inside one hospital	0.91	small, no external test
Pasini 2023 [11]	466 pts incl. LUNG1 + Radiogenomics	random 80/20 split	Acc 0.77 to 0.59	drop after ComBat
Yang 2021 [10]	645 pts, 3 centres	train one centre, test others	0.54-0.64	0.78 with random mixing
Chaunzwa 2021 [13]	311 pts, 1 hospital	external Lung3 (49 pts)	0.71 to 0.60	deep CNN
Chen 2023 [14]	402 pts, TCIA	external test (78 pts)	0.84 to 0.73	multi-task CNN
Song 2023 [12]	868 pts, 8 TCIA sets	internal + TCGA (97) + Lung3 (71)	0.82 / 0.82 / 0.80	best of 130 models, expert masks
This work	1041 pts, 4 hospitals	site-balanced CV, LOSO, 2 locked tests	0.72 / 0.71 / 0.69 / 0.59	fair metric, automatic masks, pre-registered

Table 16: Comparison with existing ADC/SCC studies (internal to external where two numbers are given)

5.3.4 Interpretation and Conclusions on Data

- **Pooled AUC can be misleading.** In multi-hospital ADC/SCC data, the hospital alone explains a pooled AUC of about 0.71. Any multi-centre result should be checked with a site-balanced metric or per-hospital AUCs.
- **There is real CT signal.** Under the fair metric the model reaches 0.722, keeps 0.714 on unseen

hospitals and 0.686 on the locked Lung3 test. The signal comes mainly from tumour position, size and the texture of the tumour border, which agrees with known clinical tendencies.

- **Harmonisation makes results trustworthy rather than higher.** ComBat removed most of the scanner fingerprint while leaving the AUC almost unchanged.
- **Data diversity mattered more than model complexity.** Adding hospitals and improving masks helped; attention, CNNs, foundation models and adversarial training did not.
- **Slice thickness is the main limit.** Texture learned mostly on thin CT does not transfer to 5 mm CT, which explains most of the external drop. The model should be used only on thin-slice CT.

5.4 Limitations of Tools (PO(e))

Table 17 lists the main technical limitations of the tools we used and their effect on the results.

Tool	Limitation	Effect / mitigation
PyRadiomics	texture depends on voxel size, bin width and slice thickness	fixed 2 mm grid and 20 HU bins; still unreliable on 5 mm CT
Resampling (SimpleITK)	cannot recover detail lost between thick slices	5 mm scans become blurred along z; reported as main failure mode
TotalSegmentator	picks the largest nodule; can miss or choose the wrong lesion	nodule found in 55/60 checks, correct lesion in about 80%; about 1 in 5 external masks may be wrong
MedSAM2	depends on the box; Dice 0.74 with loose boxes; hole filling not available on Windows	box enlarged by only 3 mm; validated on 343 expert outlines
ComBat	assumes a location-scale shift; may remove real signal when class mixes differ; needs enough patients per site	fitted per fold; new sites mapped label-free; balanced variants tested
Classifiers (LR, SVM, LightGBM)	small data, noisy labels	strong regularisation, ensembles, nested tuning
Bootstrap CI	wide intervals at n of about 80	external CIs are about plus or minus 0.12; reported in full
Hardware (4 GB GPU)	limits network size and batch size	frozen backbones; deep models restricted to last-block fine-tuning
Public labels	histology from records (NLST codes, TCGA collection)	strict inclusion rules; mixed types excluded

Table 17: Limitations of the tools used

5.5 Impact Assessment (PO(f))

5.5.1 Assessment of Societal and Cultural Issues

A reliable CT-based estimate of cancer type could reduce delays and repeat biopsies, which matters most where pathology services are limited. On the other hand, there is a risk of over-trust: a confident-looking score could be taken as a diagnosis. Communication to doctors must therefore stress that the tool supports and does not replace biopsy. Since our data contain no South Asian patients, local validation is needed to avoid a tool that works worse for the population it is meant to help.

5.5.2 Assessment of Health and Safety Issues

The pipeline is non-invasive and uses existing scans, so it creates no direct physical risk. The indirect risk is a wrong prediction. A false ADC call could delay the correct workup of an SCC patient and vice versa. With the current accuracy the model is not safe for clinical decisions. Before any clinical

use it would need prospective testing, a radiologist check of the detected nodule, and rules for when the model must not be used (for example 5 mm CT).

5.5.3 Assessment of Legal Issues

All datasets are public and de-identified, and were used under their licences: TCIA collections under CC BY 3.0/4.0 or CC BY-NC 3.0 (non-commercial, for example Lung3) and NLST data through the NCI Imaging Data Commons. The software tools are open source (Apache 2.0, BSD or MIT). A clinical version would be software as a medical device and would need regulatory approval (for example from the Directorate General of Drug Administration in Bangladesh, the FDA in the USA or CE marking in Europe), as well as compliance with patient data protection rules when used on hospital data.

5.6 Sustainability Evaluation (PO(g))

Technical sustainability. All configurations, random seeds, splits, feature names and package versions are saved, extraction is resumable, and every experiment can be rerun from saved features. The model is small and can be retrained in minutes when a new hospital joins. **Economic sustainability.** The pipeline uses free, open-source software and runs on a mid-range laptop, so a hospital would need no special hardware (Section 8). **Environmental sustainability.** Energy use is low (a few kWh for the whole project) because we avoided training large networks and reused public data. **Long-term use.** New hospitals can be added without their labels through label-free ComBat mapping, and the locked-test protocol can be repeated each time the model is updated.

5.7 Ethical Issues (PO(h))

We applied the following ethical principles:

- **Privacy:** only de-identified public data were used; no attempt was made to identify any patient, and data were stored locally and not redistributed.
- **Honest evaluation:** every fitted step was kept inside the training folds, the external tests were pre-registered, frozen with a SHA-256 hash and scored once, and post hoc analyses are clearly labelled.
- **Reporting negative results:** failed approaches (attention-MIL, foundation models, adversarial fine-tuning) and the weak TCGA result are reported in full rather than hidden.
- **Fairness:** the site-balanced metric gives every hospital equal weight, and per-hospital results are shown so that weak performance in one group is visible.
- **No overclaiming:** the model is presented as research decision support, not as a diagnostic device.
- **Credit:** all datasets, tools and published methods are cited.

One ethical challenge was the temptation to adjust the model after seeing the first external result (TCGA). We did not change the model; instead the second test (Lung3) was pre-registered with the same frozen model, and the thin/thick analysis was declared in advance for it.

6 Reflection on Individual and Team work (PO(i))

This was an individual project: Group 13 has one member. The reflection below therefore describes how the work was organised and how feedback from the instructors was used.

6.1 Individual Contribution of Each Member

Member	Contribution
Rahib Mahasin (2106119)	Topic selection and literature review; data download and preprocessing; segmentation and feature extraction; design of the fair metric and validation; all model experiments; locked external tests; figures, presentation slides and this report.

Table 18: Individual contribution

6.2 Mode of TeamWork

Work was planned weekly with a short list of goals and a written log of every experiment (phase reports with commands, results and warnings). Feedback from the instructors at the progress presentation in Week 10 (more data, external validation, clearer evaluation) directly shaped the second half of the project. Decisions that could bias the results, such as the final model and the external test rules, were written down before the corresponding data were analysed.

6.3 Diversity Statement of Team

As a single-member team, diversity came from the sources used rather than from team members: data from three countries and four clinical settings (radiotherapy, surgery, diagnosis and screening), methods from medical imaging, statistics and deep learning, and published work from many groups. The site-balanced evaluation was chosen so that smaller hospitals and minority classes are represented fairly in the result.

6.4 Log Book of Project Implementation

The project started in the mid-term break (after Week 7) and ran until the final demonstration in Week 14. Table 19 summarises the weekly log.

Date	Milestone achieved	Individual Role	Team Role	Comments
Mid-break	Topic chosen; literature review; LUNG1 data and ROI products checked; method plan locked; 29,184 local patches extracted	Literature, data preparation	Sole member	planning and setup
Week 8	Mean vs attention MIL on Gradient ROI (0.604); GTV + rim bags, 37,464 patches (0.577)	Model training	Sole member	attention unstable
Week 9	mRMR/LASSO selection (0.593); ResNet-18 patch embeddings (0.615); CLAM loss; global radiomics	Experiments	Sole member	deep features first helped attention
Week 10	Radiogenomics added (343 pts, 0.675); hospital detector 0.998; progress presentation	Data, presentation	Sole member	feedback: more data, external test
Week 11	Shortcut measured; site-balanced AUC; ComBat; shells and position features (0.681)	Method redesign	Sole member	first honest result
Week 12	Lung-PET-CT-Dx (0.718) and NLST added: 1041 pts, LOSO 0.704; TCGA downloaded and locked	Data scale-up	Sole member	about 80 GB downloaded
Week 13	MedSAM2 masks (0.722); CT-FM and adversarial tests; locked TCGA (0.585) and Lung3 (0.686)	Segmentation, testing	Sole member	pre-registered, scored once
Week 14	Slides, final demonstration and report	Communication	Sole member	project finished

Table 19: Log book of project implementation

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

Can a CT scan tell which type of lung cancer a patient has? A BUET student project has built a computer program that reads a routine chest CT scan and estimates whether a lung tumour is adenocarcinoma or squamous cell carcinoma, the two main types, which are treated differently. The program finds and outlines the tumour by itself, measures its position and the texture around it, and removes differences caused by different scanners. Tested on 1041 patients from hospitals in three countries, and then on two new datasets kept hidden until the end, it was right in about seven out of ten comparisons. It is a research tool that may one day support, not replace, biopsy.

7.2 User Manual

The steps below run the frozen model on a new CT scan. Commands are run from the project folder with the Python environments described in Section 4.

1. **Install.** Create the three environments (main, MedSAM2, TotalSegmentator) from the saved requirement files, and download the MedSAM2 checkpoint (it is loaded with `weights_only=True`).
2. **Prepare the scan.** Put each patient's CT DICOM series in its own folder. The scan should be a pre-treatment chest CT, preferably with slices of 2.5 mm or thinner.
3. **Convert and resample.** Run the build script to convert DICOM to NIFTI and resample to 2 mm.
4. **Find the tumour.** Run the detection script (TotalSegmentator lung nodules). Check the detected box on the saved preview image; if the wrong nodule was chosen, replace the box by hand.
5. **Outline the tumour.** Run the MedSAM2 script with the saved settings; it writes one mask per patient.
6. **Extract features.** Run the feature script for radiomics (shells) and position; age and sex are read from the DICOM header.
7. **Predict.** Run the prediction script. It fits the frozen model on the 1041 development patients, maps the new scans as a new site with label-free ComBat, and writes one score per patient.
8. **Read the result.** A score above 0 suggests ADC and below 0 suggests SCC; values near 0 are uncertain. The result must be interpreted by a doctor together with other clinical information.

7.3 Github Link

The complete source code, configuration files and documentation are kept in the project repository (folder ADC_SCC_MIL, sub-folders src, config, reports and docs). The core code is listed in Section 3.7.

Repository: https://github.com/<to-be-added>/ADC_SCC_CT

8 Project Management and Cost Analysis (PO(k))

This project is software only, so its costs are computing, data transfer and time rather than hardware components. The figures below are estimates in Bangladeshi Taka (BDT).

8.1 Bill of Materials

Item	Quantity / use	Cost (BDT)
Laptop with RTX 3050 GPU (already owned)	share of value used over 2 months (about 90,000 BDT over 4 years)	3,750
Storage for data (part of a 1 TB disk)	about 150 GB	900
Internet data	about 90 GB downloaded over 2 months	2,500
Electricity	about 15 kWh at about 12 BDT/kWh	180
Software (Python, PyRadiomics, scikit-learn, LightGBM, MedSAM2, TotalSegmentator)	open source	0
Datasets (TCIA, IDC)	public	0
Total		about 7,330

Table 20: Bill of materials (estimated)

8.2 Calculation of Per Unit Cost of Prototype

The prototype is the trained pipeline. Its development cost is the total of Table 20, about 7,330 BDT, plus the student's time (about 8 weeks). The compute part is small: development runs used roughly 100-150 GPU-hours of laptop time at about 0.1 kW, i.e. 10-15 kWh.

8.3 Calculation of Per Unit Cost of Mass-Produced Unit

For a deployed tool the relevant unit is one patient analysed. Measured times on the laptop were about 2.3 minutes per patient for nodule detection (3.2 hours for 85 TCGA scans), about 2 seconds for MedSAM2 (38 minutes for 1041 patients), about 2 seconds for feature extraction and less than a second for prediction, so about 2.5 minutes in total. With a power draw of 0.1 kW and electricity at 12 BDT/kWh, the cost per patient is

$$C_{\text{patient}} = 0.1 \times \underbrace{\frac{2.5}{60}}_{\text{energy}} \times 12 + \underbrace{\frac{150000}{5 \times 20000}}_{\text{workstation}} \approx 0.05 + 1.5 \approx 1.6 \text{ BDT} \quad (27)$$

assuming a dedicated 150,000 BDT workstation used for 20,000 scans per year for five years. Even with maintenance and staff time added, the cost per patient is a very small fraction of the cost of a biopsy.

8.4 Timeline of Project Implementation

Figure 23 shows the timeline as a Gantt chart. The work was done in eight weeks: the mid-term break and Weeks 8-14. The progress presentation was in Week 10 and the final demonstration in Week 14.

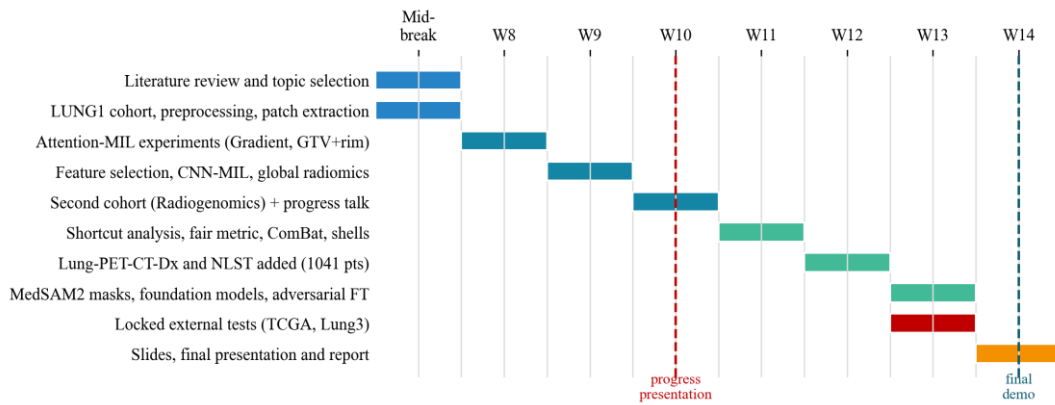


Figure 23: Gantt chart of the project.

9 Future Work (PO(I))

The project is finished, but the results point to clear next steps:

- **Robustness to thick slices.** Train with thin scans artificially blurred to 5 mm, or use features that are less sensitive to slice thickness, since 5 mm CT was the main failure mode.
- **Larger external test.** Test on a multi-hospital dataset in which every hospital has both ADC and SCC, so that per-hospital AUCs can be computed, ideally including Bangladeshi hospitals.
- **Human check of the detected nodule.** A short radiologist confirmation of the automatic box would remove most of the wrong-lesion errors (about 1 in 5).
- **PET and clinical data.** Adding PET uptake and smoking history, where available, may add information that CT texture does not contain.
- **Better use of deep models.** Foundation models could be revisited with harmonisation built into training, or with much larger multi-hospital data.
- **Prospective evaluation.** Before any clinical use, the frozen model should be tested prospectively on new patients with the same locked protocol.

10 References

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